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_DC Meter Development Pages

This contains the pages that are used during development that only have transient information which is expected to become out of date. i.e. Bugs/version faults/testing notes

2. Referenced Documents

The following table lists documents referenced within the DC Meter Firmware Design document.

ID	Reference
R1	**VDE AR E 2418 3 100 draft E** Latest draft (as of 16 Oct 2018) of the system requirements from the German authorities (PTB) used for requirements on the DC Meter.
R2	**BSI TR-03116-3** Technical Guideline BSI TR-03116 Cryptographic specifications for projects of the Federal government, Part 3: intelligent measuring systems. Referenced by the VDE AR E 2418 3 100 draft E document.
R3	**AIS 20/31** A proposal for: Functionality classes for random number generators.
R4	**ISO 18031** Information technology -- Security techniques -- Random bit generation.
R5	**NIST SP 800-90A** NIST Special Publication 800-90A Revision 1 Recommendation for Random Number Generation Using Deterministic Random Bit Generators

4. Requirements Traceability Matrix

General Requirements

VDE Requirement ID	Software Requirement Specification	Software/Firmware Design
LME001		

Key	Summary	Description
TDM-241	Display primitive error if meter knows it has failed unrecoverably	display "Err" if there is a fault with the meter. Tritium can log in to find details of the error or via technician access if available
TDM-243	session authentication id	display the unique id generated by the dc meter that identifies this charge session. This will be sent to Transparenz so that it can be mapped to the user.This unique id must also be shared with the charger so it can be used in the billing process.The session id will be created upon trigger of 'start-of-charge'. This could be current flow or an interrupt from the charger.
TDM-239	Charge time duration	Time in seconds for the charge duration. Must be with 1% of nominal value. Duration will be displayed in the format: [h]hh:mm:ss
TDM-242	comms between display and meter security status	display "SEC" if the encrypted channel is valid, "XXX" if there is an issue.
TDM-240	Meter firmware hash	Unique identifier of this release of software, which will be a 16 byte SHA2 hash of the binary. This hash can be stored at the very beginning of the binary and ignored when performing the hash calculation.
TDM-238	Meter Time Health Status	Display next to the meter time either as a symbol of confidence of accuracy of the meter clock: 1. successfully synchronised (within time period) 2. unsynchroised (hasn't sync'd since max sync duration (see config)) 3. invalid (time was sync'd but is out by 3%) 4. erred (time deviation >= 75 minutes)
TDM-231	Communications between DC Meter and remote display panel (IONITY Demo)	Ethernet to SPI bus to display module, with encryption, dedicated key, ack from display to check comms status, allows display to report errors to meter to forward to Tritium and audit log.To reduce work on FPGA, send a series of fixed-width strings always in the same order. FPGA knows how and where to display each string. Messages to the display are sent every second to keep the kWh view up to date.
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-235	Energy usage for charge session	1. Displayed in kWh (Appendix B suggests could optionally use MWh but [ERA012] says must be kWh) 2. The format of the usage will be: `[0-9]{7}.[0-9]{2}` e.g. 0000000.00
TDM-167	Event Log	Provides persistent Logging functions for Meter.
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.

11 issues

LME002

Key	Summary	Description
TDM-241	Display primitive error if meter knows it has failed unrecoverably	display "Err" if there is a fault with the meter. Tritium can log in to find details of the error or via technician access if available
TDM-243	session authentication id	display the unique id generated by the dc meter that identifies this charge session. This will be sent to Transparenz so that it can be mapped to the user.This unique id must also be shared with the charger so it can be used in the billing process.The session id will be created upon trigger of 'start-of-charge'. This could be current flow or an interrupt from the charger.
TDM-239	Charge time duration	Time in seconds for the charge duration. Must be with 1% of nominal value. Duration will be displayed in the format: [h]hh:mm:ss
TDM-242	comms between display and meter security status	display "SEC" if the encrypted channel is valid, "XXX" if there is an issue.
TDM-240	Meter firmware hash	Unique identifier of this release of software, which will be a 16 byte SHA2 hash of the binary. This hash can be stored at the very beginning of the binary and ignored when performing the hash calculation.
TDM-238	Meter Time Health Status	Display next to the meter time either as a symbol of confidence of accuracy of the meter clock: 1. successfully synchronised (within time period) 2. unsynchroised (hasn't sync'd since max sync duration (see config)) 3. invalid (time was sync'd but is out by 3%) 4. erred (time deviation >= 75 minutes)
TDM-231	Communications between DC Meter and remote display panel (IONITY Demo)	Ethernet to SPI bus to display module, with encryption, dedicated key, ack from display to check comms status, allows display to report errors to meter to forward to Tritium and audit log.To reduce work on FPGA, send a series of fixed-width strings always in the same order. FPGA knows how and where to display each string. Messages to the display are sent every second to keep the kWh view up to date.
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-235	Energy usage for charge session	1. Displayed in kWh (Appendix B suggests could optionally use MWh but [ERA012] says must be kWh) 2. The format of the usage will be: `[0-9]{7}.[0-9]{2}` e.g. 0000000.00
TDM-167	Event Log	Provides persistent Logging functions for Meter.
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.

11 issues

LME003

Key	Summary	Description
TDM-231	Communications between DC Meter and remote display panel (IONITY Demo)	Ethernet to SPI bus to display module, with encryption, dedicated key, ack from display to check comms status, allows display to report errors to meter to forward to Tritium and audit log.To reduce work on FPGA, send a series of fixed-width strings always in the same order. FPGA knows how and where to display each string. Messages to the display are sent every second to keep the kWh view up to date.
TDM-235	Energy usage for charge session	1. Displayed in kWh (Appendix B suggests could optionally use MWh but [ERA012] says must be kWh) 2. The format of the usage will be: `[0-9]{7}.[0-9]{2}` e.g. 0000000.00
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.

3 issues

Charging point measuring device with electricity metering function

VDE Requirement ID	Tritium Design Element ID

LME010

Key	Summary	Description
TDM-186	calibration table(s)	Used to convert readings from the DC Sensor to be accurate. At the moment this is: 1. offset 2. scale factor readings will be store as: raw value, offset, scaling factor, calculated result so original reads can be re-calculated if it was later determined the offset/scaling was incorrect.

1 issue

LME011

Key	Summary	Description
TDM-186	calibration table(s)	Used to convert readings from the DC Sensor to be accurate. At the moment this is: 1. offset 2. scale factor readings will be store as: raw value, offset, scaling factor, calculated result so original reads can be re-calculated if it was later determined the offset/scaling was incorrect.

1 issue

LME012

Key	Summary	Description

No issues found

LME013	Key Summary Description No issues found
LME014	Key Summary Description No issues found
LME015	Key Summary Description No issues found
LME016	Key Summary Description No issues found
LME017	Key Summary Description No issues found

Charging Point Measuring Device With Time Measurement Function	
VDE Requirement ID	**Tritium Design Element ID**
LME020	Key Summary Description TDM-196 time duration calculation Based on lessons learned with time synchronisation, calculating duration is a combination of a start time and a monotonic clock counting forwards. If an end time is required it is calculated from start time + duration.This time duration capability can also be used to implement timeouts within the system. 1 issue
LME021	Key Summary Description TDM-312 Determine the number of reads/interval to calculate usage how many samples per second do we need to meet accuracy requirements. 1 issue

LME022	Key Summary No issues found
LME023	Key Summary No issues found
LME024	Key Summary TDM-239 Charge time duration TDM-231 Communications between DC Meter and remote display panel (IONITY Demo) 2 issues

Measured Value Acquisition Module	
VDE Requirement ID	**Tritium Design Element ID**
MWE001	Key Summary Description TDM-195 Format of recorded usage measurements NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 1 issue
MWE002	Key Summary Description TDM-195 Format of recorded usage measurements NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 1 issue

MWE003	Key / Summary / Description TDM-195 / Format of recorded usage measurements / NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 1 issue
MWE004	Key / Summary / Description TDM-862 / Decimal Fixed Point Numbers / See https://wiki.tritium.com.au/display/RND/Decimal+Fixed+Point+Numbers TDM-195 / Format of recorded usage measurements / NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 2 issues
MWE005	Key / Summary / Description TDM-195 / Format of recorded usage measurements / NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 1 issue

Measured Value Processing Module	
VDE Requirement ID	**Tritium Design Element ID**
MWV001	Key / Summary / Description TDM-195 / Format of recorded usage measurements / NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 1 issue

MWV002

Key	Summary	Description
TDM-195	Format of recorded usage measurements	NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2

1 issue

MWV003

Key	Summary	Description
TDM-313	Determine smoothing algorithm	Post processing of data to remove anomalous reads (or spikes) but still maintain accuracy requirements. Errors in calculations shall be captured and logged/reported as part of telemetry.
TDM-154	Usage Session Measurements Recording /Reporting	stored measurements will be "as-is" without calculation but will contain enough information with each read so that a calculation may be performed at any time.use common 'circular buffer' mechanism for session related measurements
TDM-312	Determine the number of reads/interval to calculate usage	how many samples per second do we need to meet accuracy requirements.
TDM-160	Find Accurate Arithmetic Library	There are requirements regarding the accuracy of any calculation, which add pressure on the FPU because it inherently has error. So to reduce issues with certification a library that is more accurate needs to be found and used. The requirements regarding accuracy are: > *[MWV004] When calculating derived measurements, DO NOT add an error contribution of more than ± 1% of the least significant used digit of the calculated value.* > *[MWV005] A set of derived measurements may not degrade the resolution of the measurements obtained from the MWE component.*
TDM-195	Format of recorded usage measurements	NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2

5 issues

MWV005

Key	Summary	Description
TDM-154	Usage Session Measurements Recording/Reporting	stored measurements will be "as-is" without calculation but will contain enough information with each read so that a calculation may be performed at any time.use common 'circular buffer' mechanism for session related measurements
TDM-160	Find Accurate Arithmetic Library	There are requirements regarding the accuracy of any calculation, which add pressure on the FPU because it inherently has error. So to reduce issues with certification a library that is more accurate needs to be found and used. The requirements regarding accuracy are: *[MWV004] When calculating derived measurements, DO NOT add an error contribution of more than ± 1% of the least significant used digit of the calculated value.* *[MWV005] A set of derived measurements may not degrade the resolution of the measurements obtained from the MWE component.*
TDM-862	Decimal Fixed Point Numbers	See https://wiki.tritium.com.au/display/RND/Decimal+Fixed+Point+Numbers

3 issues

Measured Value Memory

VDE Requirement ID	Tritium Design Element ID

STORAGE001

Key	Summary	Description
TDM-153	UID generation for charge sessions	independent id. Maps to user somehow
TDM-195	Format of recorded usage measurements	NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2

2 issues

STORAGE002

Key	Summary	Description
TDM-153	UID generation for charge sessions	independent id. Maps to user somehow
TDM-195	Format of recorded usage measurements	NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2

2 issues

STORAGE003	<table><tr><th>Key</th><th>Summary</th><th>Description</th></tr><tr><td>TDM-195</td><td>Format of recorded usage measurements</td><td>NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2</td></tr></table> 1 issue
STORAGE004	<table><tr><th>Key</th><th>Summary</th><th>Description</th></tr><tr><td>TDM-195</td><td>Format of recorded usage measurements</td><td>NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2</td></tr></table> 1 issue
STORAGE005	<table><tr><th>Key</th><th>Summary</th><th>Description</th></tr><tr><td>TDM-197</td><td>packaging stored measurements into a SAFE Initiative payload</td><td>Using OCMF instead of SML due to time constraints and licensing issues with libraries.</td></tr></table> 1 issue

Time Management Method	
VDE Requirement ID	**Tritium Design Element ID**

TIME001

Key	Summary	Description
TDM-208	last time sync time	
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

3 issues

TIME002

Key	Summary	Description
TDM-238	Meter Time Health Status	Display next to the meter time either as a symbol of confidence of accuracy of the meter clock: 1. successfully synchronised (within time period) 2. unsynchroised (hasn't sync'd since max sync duration (see config)) 3. invalid (time was sync'd but is out by 3%) 4. erred (time deviation >= 75 minutes)
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

2 issues

TIME003

Key	Summary	Description
TDM-209	battery/supercap voltage (RTC and device)	
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

3 issues

TIME004

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

TIME005

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

TIME006

Key	Summary

No issues found

TIME007

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

TIME008

Key	Summary	Description
TDM-175	check RTC value	Tests to see if RTC value is approximately accurate. There will be the ability to store the current RTC value(s) when main power disappears. When power returns the stored (and time sync hasn't happened yet) time should be older than current time. Otherwise the RTC was reset completely.
TDM-174	check RTC battery /supercap voltage	TBD if using external RTC chip (with higher accuracy).1. communicate with RTC with read/write/interrupts2. react to battery bad flag on RTC
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

3 issues

TIME009

Key	Summary	Description
TDM-174	check RTC battery /supercap voltage	TBD if using external RTC chip (with higher accuracy).1. communicate with RTC with read/write/interrupts2. react to battery bad flag on RTC
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

2 issues

TIME010

Key	Summary	Description
TDM-238	Meter Time Health Status	Display next to the meter time either as a symbol of confidence of accuracy of the meter clock: 1. successfully synchronised (within time period) 2. unsynchroised (hasn't sync'd since max sync duration (see config)) 3. invalid (time was sync'd but is out by 3%) 4. erred (time deviation >= 75 minutes)
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

2 issues

TIME011

Key	Summary	Description

No issues found

TIME012

Key	Summary	Description
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.

1 issue

TIME013

Key	Summary	Description
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.

1 issue

TIME014

Key	Summary	Description
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.

1 issue

TIME015

Key	Summary	Description
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

TIME016

Key	Summary	Description
TDM-184	Configuration of time	Time is automatically set as soon as the DC Meter has connection to a listed clock source. The configuration includes: • details of clock sources, • timezone, • time synchronisation frequency (specified in minutes, min 2 minutes [TIME012], max value = 7days (60*24*7) [TIME013]). • max time between sync before panicking When time is displayed it has to represent timezone and daylight savings, but when time is recorded, it SHALL be UTC.
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two-part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)
2 issues		

TIME017	

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

2 issues

Time Server	
VDE Requirement ID	**Tritium Design Element ID**

LOG001

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

2 issues

LOG002

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record

1 issue

LOG003

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record

1 issue

LOG004

Key	Summary	Description
TDM-155	Event Log Recording /Reporting	use common 'circular buffer' mechanism. The size of the event log must be large enough to store (8 + 2) years of events

1 issue

LOG005

Key	Summary	Description
TDM-230	Technician Access (VDE Demo)	A technician will require access to the DC meter. The DC meter has an Ethernet connection and is already required to have TLS 1.2 client functionality. • Implement TLS 1.2 server with required client side certificate authorisation. • Client will present to the server a certificate to prove it is approved to access the server. ~~Requires confirmation from James on direction of product. Need to know if we are expected to store cost on tariffs as a technician role could be to update tariff info. Technicians may perform a test locally on the meter via PC /Laptop login over local network to verify measurement accuracy with varying loads, so a live feed of usage (updating every 5 seconds???).~~

1 issue

LOG006

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record

1 issue

LOG007

Key	Summary	Description
TDM-155	Event Log Recording /Reporting	use common 'circular buffer' mechanism. The size of the event log must be large enough to store (8 + 2) years of events

1 issue

LOG008

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG009

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG010

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG011

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG012

Key	Summary	Description

No issues found

LOG013

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG014

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG015

Key	Summary	Description
TDM-183	DC Meter Run Time Self Test	Slow periodic tests performed every 6 hours. Continuous tests performed at variable rates depending on test (~30sec) Self test performed by supervisory state machine whilst not charging. Results of all tests need to be logged to the event log. Tests not presently contained in this task: • Check communications interfaces • MQTT connection • NTP connection • TSO connection • AFE comms • Flash comms Bootloader image integrity check (SHA256) is completed before jumping to it upon a FW update request
TDM-148	DC Meter Power On Self Test	Power up self-test run by supervisory state machine. Either runs or ensures result was valid of tests. Results of all tests need to be logged to the event log. • User image integrity check (SHA256) is completed by bootloader before jumping to it (Otherwise a FW update is attempted or BL holds operation of device)
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

3 issues

LOG016

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)

1 issue

LOG017

Key	Summary	Description
TDM-223	statistics	1. max size (bytes) 2. max size (records) 3. used (bytes and records, as a %) 4. bad record count (fails hash test) 5. first record timestamp 6. last record timestamp 7. Thanks to WELMEC7.2:P7, may need to record/report a counter for number of times any configuration field was changed
TDM-226	The circular buffer shall provide notification events	notify listeners of events like: 1. size usage (75%, 80%, 90%, 100%) 2. reset/empty/delete/overwrite
TDM-155	Event Log Recording/Reporting	use common 'circular buffer' mechanism. The size of the event log must be large enough to store (8 + 2) years of events

3 issues

LOG018

Key	Summary	Description
TDM-234	share random user id for charge session with charger/CPO	
TDM-223	statistics	1. max size (bytes) 2. max size (records) 3. used (bytes and records, as a %) 4. bad record count (fails hash test) 5. first record timestamp 6. last record timestamp 7. Thanks to WELMEC7.2:P7, may need to record/report a counter for number of times any configuration field was changed
TDM-167	Event Log	Provides persistent Logging functions for Meter.

3 issues

LOG019

Key	Summary	Description
TDM-223	statistics	1. max size (bytes) 2. max size (records) 3. used (bytes and records, as a %) 4. bad record count (fails hash test) 5. first record timestamp 6. last record timestamp 7. Thanks to WELMEC7.2:P7, may need to record/report a counter for number of times any configuration field was changed

1 issue

LOG020

Key	Summary	Description
TDM-167	Event Log	Provides persistent Logging functions for Meter.

1 issue

LOG021

Key	Summary	Description
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

1 issue

LOG022

Key	Summary	Description
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

1 issue

LOG023

Key	Summary	Description
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record

1 issue

LOG024

Key	Summary	Description
TDM-225	record hashing the buffer internals (pointers, stats, etc)	To help detect if the buffer management becomes corrupt, we hash the internal variables used for management of records.
TDM-224	record hashing	Each record contains it's own hash to detect corruption. The hash includes all data fields.
TDM-198	Event Log Format	1. record index (uint32) 2. timestamp (local time, unix sec uint64_t) 3. SHA hash to verify record integrity : uint8_t[32] 4. event type id : uint8_t 5. old value/new value/description : uint8_t[83] total size 128bytes per record

3 issues

LOG025

Key	Summary	Description

No issues found

LOG026

Key	Summary	Description
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

1 issue

LOG027

Key	Summary	Description
TDM-230	Technician Access (VDE Demo)	A technician will require access to the DC meter. The DC meter has an Ethernet connection and is already required to have TLS 1.2 client functionality. • Implement TLS 1.2 server with required client side certificate authorisation. • Client will present to the server a certificate to prove it is approved to access the server. ~~Requires confirmation from James on direction of product. Need to know if we are expected to store cost on tariffs as a technician role could be to update tariff info. Technicians may perform a test locally on the meter via PC/Laptop login over local network to verify measurement accuracy with varying loads, so a live feed of usage (updating every 5 seconds???).~~
TDM-245	Technician Access messages to export/report parts of data	The private keys shall never be extracted or viewed. Public keys can be extracted or viewed if permissions permit. The state of the meter can be viewed. The CPO can export data. Technicians with the appropriate key can export measurements and event logs. Update live values to the Technician Access terminal
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

3 issues

LOG029

Key	Summary	Description
TDM-155	Event Log Recording /Reporting	use common 'circular buffer' mechanism. The size of the event log must be large enough to store (8 + 2) years of events
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

2 issues

LOG030	

Key	Summary	Description
TDM-222	The circular buffer can be exported	export the whole buffer, using encryption, to an external interface. A separate flag will indicate if deletion of the contents is to happen after the export. Some kind of handshake is required to ensure what was exported is valid /readable before deleting. Exporting creates an entry in the event log. Exported data must contain a header which includes: 1. meter unique identifier 2. timestamp of the export (according to the meter) 3. name of the buffer (e.g. measurements, events, etc) Once data has been exported, it is marked as exported, which allows it to be automatically overwritten with newer records when space requires. Data can still be exported if marked for deletion.

1 issue

Firmware Management Module	
VDE Requirement ID	**Tritium Design Element ID**
FWM001	

Key	Summary	Description
TDM-212	encrypted comms	~~Communicates to the Tritium server's to download updates~~
TDM-217	validate upgrade process and start metering software	no comms to charger means charger can't charge. Once metering software started, comms to charger recommences and a charge can occur.
TDM-216	validate upgrade binary	different parts of the meter can be upgraded (metering/event log can't be):1. firmware, 2. calibration tables, 3. configuration
TDM-215	validate binary provider	Only a trusted source can provide a binary upgrade. There needs to be a mechanism to authenticate the binary against the source of the binary, which is separate to the encryption of the binary.

4 issues

FWM002	

Key	Summary	Description
TDM-212	encrypted comms	~~Communicates to the Tritium server's to download updates~~
TDM-217	validate upgrade process and start metering software	no comms to charger means charger can't charge. Once metering software started, comms to charger recommences and a charge can occur.
TDM-216	validate upgrade binary	different parts of the meter can be upgraded (metering/event log can't be):1. firmware, 2. calibration tables, 3. configuration
TDM-215	validate binary provider	Only a trusted source can provide a binary upgrade. There needs to be a mechanism to authenticate the binary against the source of the binary, which is separate to the encryption of the binary.

4 issues

FWM003	

Key	Summary	Description

No issues found

FWM004

Key	Summary	Description
TDM-218	during upgrade, the display will show progress and upgrade version	requires code to communicate with the display (which should already be done). Create message to display regarding the upgrade process.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

FWM005

Key	Summary

No issues found

FWM006

Key	Summary	Description
TDM-217	validate upgrade process and start metering software	no comms to charger means charger can't charge. Once metering software started, comms to charger recommences and a charge can occur.

1 issue

FWM007

Key	Summary	Description
TDM-158	Upgrading (Bootloader)	Always runs on start up.Checks if there is a binary to install and performs necessary steps to install it safely. Always checks the validity of the software it is about to start.existing measurements/audit trail/etc must remain intact after an upgrade

1 issue

Signature Module	
VDE Requirement ID	**Tritium Design Element ID**

SIGN001

Key	Summary	Description
TDM-191	find/search for keys	to be used internally by the metering app. Private key values can't be displayed, exported, or printed.1. find/search for key by privilege/functionality2. find/search for key by user (a.k.a. public key)3. find/search for key by flag (active /invalidated/unused/private/public)4. find/search for key by id5. find/search for key based on key
TDM-185	Configuration of security key (s)	This task only addresses the basic get/set functionality. Further management needs are defined in task "Security Keys Management"

2 issues

SIGN002

Key	Summary	Description

No issues found

SIGN003

Key	Summary	Description
TDM-173	Prove random number generator	NIST have test code that can provide quality assurance on results from a random number generator. This is a combination of getting test code running and generating numbers from the hardware generator. This is an R&D task as the process for converting bits from the hardware RBG may require tuning to pass tests. 1. Download code 2. Get it to compile on Linux and run provided sample set to verify it works 3. Get it to run on target 4. Measure performance to see if we can run on target as part of run time validation (or a subset of)

1 issue

SIGN004

Key	Summary	Description

No issues found

SIGN005

Key	Summary	Description
TDM-189	unique identification	Ensure adding new keys doesn't result in a duplication of id.Removing/replacing keys deletes an id and allows it to be used again.Search algorithm to detect next available id
TDM-185	Configuration of security key(s)	This task only addresses the basic get/set functionality. Further management needs are defined in task "Security Keys Management"

2 issues

SIGN006

Key	Summary	Description
TDM-191	find/search for keys	to be used internally by the metering app. Private key values can't be displayed, exported, or printed.1. find/search for key by privilege/functionality2. find/search for key by user (a.k.a. public key)3. find/search for key by flag (active /invalidated/unused/private/public)4. find/search for key by id5. find/search for key based on key

1 issue

SIGN007

Key	Summary

No issues found

SIGN008	Key Summary No issues found
SIGN009	Key Summary No issues found
SIGN010	Key Summary Description TDM-192 invalidate key (s) by id Keeping a list of previously used pairs so they can't be reactivated by programming as if they are new TDM-191 find/search for keys to be used internally by the metering app. Private key values can't be displayed, exported, or printed.1. find/search for key by privilege/functionality2. find/search for key by user (a.k.a. public key)3. find/search for key by flag (active /invalidated/unused/private/public)4. find/search for key by id5. find/search for key based on key 2 issues
SIGN011	Key Summary Description No issues found

Verification Module	
VDE Requirement ID	**Tritium Design Element ID**
VERIFY001	Key Summary No issues found
VERIFY002	Key Summary No issues found

VERIFY003

Key	Summary	Description
TDM-172	add security library (mbed TLS) to the meter	ECDH, TLS, X.509, entropy management, etc. https://tls.mbed.org This library is capable of doing everything we want with regards to security signing, authentication, HMAC, encryption, random bit generator/entropy. However, we need to port it to the target. Some parts of the library need us to create code to fill in stubs and provide the missing functionality. We also need to performance test to see if we can still operate fast enough.

1 issue

VERIFY004

Key	Summary	Description
TDM-172	add security library (mbed TLS) to the meter	ECDH, TLS, X.509, entropy management, etc. https://tls.mbed.org This library is capable of doing everything we want with regards to security signing, authentication, HMAC, encryption, random bit generator/entropy. However, we need to port it to the target. Some parts of the library need us to create code to fill in stubs and provide the missing functionality. We also need to performance test to see if we can still operate fast enough.

1 issue

VERIFY005

Key	Summary	Description
TDM-172	add security library (mbed TLS) to the meter	ECDH, TLS, X.509, entropy management, etc. https://tls.mbed.org This library is capable of doing everything we want with regards to security signing, authentication, HMAC, encryption, random bit generator/entropy. However, we need to port it to the target. Some parts of the library need us to create code to fill in stubs and provide the missing functionality. We also need to performance test to see if we can still operate fast enough.

1 issue

VERIFY006

Key	Summary	Description
TDM-199	event id list	1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed)
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

Verification Compliant	
VDE Requirement ID	**Tritium Design Element ID**

ERA007

Key	Summary	Description
TDM-919	Display the cumulative (for life) kW register value	
TDM-152	Cumulative tariff registers summing all charge sessions	name/identifiercumulative storage, no resetting back to 0 unless counter overflow.Protection against duplicate tariffscreate/update/deletestart time (end time assumed by next tariff)
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

3 issues

ERA008

Key	Summary	Description
TDM-919	Display the cumulative (for life) kW register value	
TDM-152	Cumulative tariff registers summing all charge sessions	name/identifiercumulative storage, no resetting back to 0 unless counter overflow.Protection against duplicate tariffscreate/update/deletestart time (end time assumed by next tariff)
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

3 issues

ERA009

Key	Summary	Description
TDM-151	Tariff settings	Seasonal and/or time of day rate changes. 1. will need the fields: yyyy/mm/dd hh:mm, name, id, $/kWh, duration (minutes) 2. limit number of tariffs to 8 3. add ability to change tariffs through the terminal Further information on types of tariffs to cater for can be found in the following links (Note: these links are in German so translation is required): • https://wiki.scn.sap.com/wiki/pages/viewpage.action?pageId=446731342 • https://help.sap.com/viewer/3e88baa8ca4e4863a6203e00deb32df2/2.3.0.1/de-DE/3ba4e45461c2136de10000000a441470.html
TDM-194	maintain a cumulative measurement registers summing current charge session for each tariff	A single charging session may cover multiple tariffs. The user needs to be aware of this during the charge and end of charge.
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

4 issues

ERA010

Key	Summary	Description
TDM-151	Tariff settings	Seasonal and/or time of day rate changes. 1. will need the fields: yyyy/mm/dd hh:mm, name, id, $/kWh, duration (minutes) 2. limit number of tariffs to 8 3. add ability to change tariffs through the terminal Further information on types of tariffs to cater for can be found in the following links (Note: these links are in German so translation is required): • https://wiki.scn.sap.com/wiki/pages/viewpage.action?pageId=446731342 • https://help.sap.com/viewer/3e88baa8ca4e4863a6203e00deb32df2/2.3.0.1/de-DE/3ba4e45461c2136de10000000a441470.html
TDM-194	maintain a cumulative measurement registers summing current charge session for each tariff	A single charging session may cover multiple tariffs. The user needs to be aware of this during the charge and end of charge.
TDM-152	Cumulative tariff registers summing all charge sessions	name/identifiercumulative storage, no resetting back to 0 unless counter overflow. Protection against duplicate tariffscreate/update/deletestart time (end time assumed by next tariff)
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.
TDM-924	Can you give us an example of a charge session with multiple tariffs?	The VDE spec doesn't contain detailed description on Tariffs. There are discussions around roaming tariffs, tariffs based on time, tariffs based on usage, but nothing concrete. See: • ~~TDM-151~~ • ~~TDM-152~~ • ~~TDM-194~~ • ~~TDM-236~~
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

6 issues

ERA011

Key	Summary	Description
TDM-235	Energy usage for charge session	1. Displayed in kWh (Appendix B suggests could optionally use MWh but [ERA012] says must be kWh) 2. The format of the usage will be: `[0-9]{7}.[0-9]{2}` e.g. 0000000.00

1 issue

ERA012

Key	Summary	Description
TDM-235	Energy usage for charge session	1. Displayed in kWh (Appendix B suggests could optionally use MWh but [ERA012] says must be kWh) 2. The format of the usage will be: `[0-9]{7}.[0-9]{2}` e.g. 0000000.00
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

ERA013

Key	Summary	Description
TDM-239	Charge time duration	Time in seconds for the charge duration. Must be with 1% of nominal value. Duration will be displayed in the format: [h]hh:mm:ss
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

ERA014

Key	Summary	Description
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

ERA015

Key	Summary	Description
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

ERA016

Key	Summary	Description
TDM-237	Meter Time	1. Display local time. This means including DST and Timezone values. 2. The Time displayed SHALL update every second 3. Time SHALL be sent to the display as a string 4. Time SHALL be displayed in the format yyyy-mm-ddTHH:mm:ss z/+hh:mm
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

2 issues

ERA017

Key	Summary	Description
TDM-241	Display primitive error if meter knows it has failed unrecoverably	display "Err" if there is a fault with the meter. Tritium can log in to find details of the error or via technician access if available
TDM-167	Event Log	Provides persistent Logging functions for Meter.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. ****REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS****

3 issues

ERA018

Key	Summary	Description
TDM-242	comms between display and meter security status	display "SEC" if the encrypted channel is valid, "XXX" if there is an issue.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

ERA019

Key	Summary	Description
TDM-242	comms between display and meter security status	display "SEC" if the encrypted channel is valid, "XXX" if there is an issue.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

ERA020

Key	Summary	Description
TDM-194	maintain a cumulative measurement registers summing current charge session for each tariff	A single charging session may cover multiple tariffs. The user needs to be aware of this during the charge and end of charge.
TDM-236	Active Tariff	Tariffs can be used in multiple ways. Seasonal, time-of-day, or a combination. When not charging display the active tariff. When charging display the tariffs used so far during the charge period.

2 issues

ERA021

Key	Summary	Description
TDM-243	session authentication id	display the unique id generated by the dc meter that identifies this charge session. This will be sent to Transparenz so that it can be mapped to the user.This unique id must also be shared with the charger so it can be used in the billing process.The session id will be created upon trigger of 'start-of-charge'. This could be current flow or an interrupt from the charger.

1 issue

ERA022

Key	Summary	Description
TDM-238	Meter Time Health Status	Display next to the meter time either as a symbol of confidence of accuracy of the meter clock: 1. successfully synchronised (within time period) 2. unsynchroised (hasn't sync'd since max sync duration (see config)) 3. invalid (time was sync'd but is out by 3%) 4. erred (time deviation >= 75 minutes)
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

ERA023

Key	Summary	Description
TDM-243	session authentication id	display the unique id generated by the dc meter that identifies this charge session. This will be sent to Transparenz so that it can be mapped to the user.This unique id must also be shared with the charger so it can be used in the billing process.The session id will be created upon trigger of 'start-of-charge'. This could be current flow or an interrupt from the charger.
TDM-1004	What info needs to be on the remote display	Sub tasks of this Story are individual questions to be asked of the VDE certification house that could relate to design, implementation, or test strategies. This way each question manages its own responses. The use of labels is encouraged to help as filters on related questions when searching or documenting via the Wiki. **REMEMBER TO ADD THE ITEMS AS TASKS. NOT SUB-TASKS**

2 issues

ERA024

Key	Summary	Description
TDM-179	check communications	
TDM-183	DC Meter Run Time Self Test	Slow periodic tests performed every 6 hours. Continuous tests performed at variable rates depending on test (~30sec) Self test performed by supervisory state machine whilst not charging. Results of all tests need to be logged to the event log. Tests not presently contained in this task: • Check communications interfaces • MQTT connection • NTP connection • TSO connection • AFE comms • Flash comms Bootloader image integrity check (SHA256) is completed before jumping to it upon a FW update request
TDM-190	different security keys for different functionality	A method to authenticate a user to a set of privileges/functionality. A separate key for has.to.be, Tritium, technician, etc.
TDM-231	Communications between DC Meter and remote display panel (IONITY Demo)	Ethernet to SPI bus to display module, with encryption, dedicated key, ack from display to check comms status, allows display to report errors to meter to forward to Tritium and audit log.To reduce work on FPGA, send a series of fixed-width strings always in the same order. FPGA knows how and where to display each string. Messages to the display are sent every second to keep the kWh view up to date.
TDM-185	Configuration of security key(s)	This task only addresses the basic get/set functionality. Further management needs are defined in task "Security Keys Management"
TDM-244	Client authorisation for access to the DC Meter	Anyone attempting access to the DC Meter as a "client" (i.e. the DC Meter is the server) to use the terminal interface of the DC Meter first is authenticated using X.509 certificates that were created by Tritium. An example is the technician must authorise themselves with the meter before seeing the menu and retrieving information. The CPO must authorise prior to updating tariffs.
TDM-973	Verification Compliancy	ERA024 - To what standard? Is TR-03107-1 an acceptable standard?

7 issues

User Interface

VDE Requirement ID	Tritium Design Element ID
UI001	Key: TDM-248 — Summary: Terminal input validation of write commands for range testing and privileges for the user — Description: 1 issue
UI002	Key — Summary — Description No issues found

Authorization Unit

VDE Requirement ID	Tritium Design Element ID
AUTHZ001	Key Summary No issues found
AUTHZ002	Key / Summary / Description TDM-195 — Format of recorded usage measurements — NOTE: our LME (charge point measureing device) is self contained so [MWE001] is not applicable. In our architecture we should always be capable of measuring otherwise we have a significant problem that means we don't operate, so using a wild card as per [MWE005] doesn't apply.A separate entry for each tariff that was used during the charge. See the following link for details of data stored: https://wiki.tritium.com.au/download/attachments/41746575/StaticMemoryRequirements.xlsx?api=v2 TDM-199 — event id list 1. startup (type of reset) 2. power loss 3. ~~power drop~~ 4. charge session started (EDSI) 5. charge session ended (EDSI + charger session ID) 6. configuration parameter change ~~(+ parameter name + old + new - 128 byte memory structure written as two part 64 byte data events)~~ (+chunk number + status of the setting new data: 1-success, 0-fail (only in first chunk) +new param data as data events) a. AFE b. TSOIF c. PCBA TEST LOG d. SUBA TEST LOG e. HOST PRIVATE KEY f. TRUST ROOT KEY 7. ~~invalid measurement due to time (+ invalid time, range start + end) [covered by below]~~ 8. Out of Service during charge (+Out Of Service reason) 9. ~~invalid measurement due to hash mismatch (entry timestamp)~~ 10. time synchronisation (SUCCESS/FAIL) 11. ~~time failure (e.g. time > 3% legal time)~~ 12. time deviation > 75 minutes 13. RTC battery fail 14. time between sync > safe period 15. ~~suspect faulty time source~~ 16. ~~POST/RTST results~~ 17. technician login/logout ~~and any commands used~~ (+login/logout +permission level) 18. ~~tamper detection~~ 19. log export occurred (+type log) 20. firmware update download (+success/fail +upgrade server trustworthiness +image trustworthiness +image hash) 21. ~~firmware update install (success/fail)~~ 22. (~~de~~)commission 23. measurement storage usage warning (75% usage, 80%, 90%, 100%) 24. ~~request for data (historical)~~ 25. ~~records marked for deletion (old enough and have been exported)~~ 26. records deleted (logical address block) 27. error on signature validation (+signature interface) [not TSO 28. error on signature preparation (+signature interface) [not TSO] 29. self test failure (+test failed) 2 issues

Authentication Unit	
VDE Requirement ID	**Tritium Design Element ID**

AUTHN001	Key Summary No issues found
AUTHN002	Key Summary No issues found
AUTHN003	Key Summary No issues found
AUTHN004	Key Summary No issues found

5. Inverse Traceability Matrix

Firmware and Software:

Key	Summary	T	Created	Updated	Due	Assignee	Reporter	P	Status	Resolution	Linked Issues

No issues found

DCM Box Testing

Following tests started from M2 hardware with firmware without calibration (30/09/2019).

Fault/Behaviour	Cause	Detailed Description	Closed
Removal of measured voltage does not stop charge session	ChargeSessionSM.cpp logic doesn't check this	Once a charge is started above current/voltage thresholds; voltage can be removed and charge session doesn't stop. Current being removed successfully stops session.	•
Enter Out of Service (OoS) mode	Underflow on power calculation	OoS cause reported as READ_ADC_FAIL. Goes OoS when no power is 0.000.	•
Energy displays ??? not value		Both debug Ws and TTI kWh display ??? instead of values. Occasionally a 'blip' will occur and what looks like power is shown for 1s.	•
TTI Energy displays Ws in a given loop not session energy	Temporary Ws shown	Needs to be total session energy	•
Doesn't support negative power. Rolls back total energy		Negative voltage doesn't latch in state machine. Energy counter rolls back when negative power is applied. Is this a problem? Does negative energy need independent counter	•
Negative current does not display correctly		AFE current shows negative but output current/debug current shows absolute.	•
Debug print of power shows VVVVV.VV when above 100kW	String print digit length	Does not have enough digits for power/Ws/voltage	•
Cable compensation voltage change can't be seen	Initialisation order	Voltage measured isn't changed by current when cable resistance is written to as expected.	•
Cable compensation seems to be based off opposite cable	Hardware abstraction opposite to labelled	Uses resistance value written to opposing cable	•
Uncertainty of sufficient energy transferred by IR LED	Non-standard test equipment	IR LED energy received is difficult to determine based on non-ideal receiver. Changed resistor to increase drive current.	•
AFE no longer reads/displays negative voltage /current	Typecast when calibrated to uint	Seems to have changed when calibration was introduced	•
Voltage calibration is the same for both voltage sense inputs		Calibration changing added to cable switching	•
Configuration isn't loaded once unit is reset		Not written and/or read in memory.	•
Temperature sensor reads false values at negative temps	Typecast to uint	Read ~6516 at negative temperatures	•
At -40'C one cable sense being shorted effects the measurement of the other (only shorting cable 2 with cable 1)	Leakage through pull down	Both sets of opto switches are turning on.	•
Bootloading resets the calibration information	Any JSON sent info writes to memory.	Changed so bootloading command does not	•

Non-issues/Accepted for first release

OoS doesn't recover		No OoS recovery mechanism yet	
Measurements displays ??? when out of range		E.g. If over 600A does not display the reading - should this be changed to maximum possible measurement rather than maximum expected? This is expected behaviour	
If no internet connection on startup cannot connect to TTI even on restore of internet		Unit powered up networked to router without external internet connectivity. Attempt to connect to TTI, fails as expected. Restore internet connectivity and attempt to connect to TTI, still fails.	
Charge start time/duration don't appear on TTI		Not yet implemented	
Whilst bootloading TTI must be closed or process crashes		Not yet fixed	

DC meter - overview of stories

All stories

Key	Summary	T	Updated	Assignee	Status	wp	Labels	Linked Issues
TDM-160	Find Accurate Arithmetic Library		Jul 01, 2019	Jeremy Kincaid	DONE	1841	accuracy investigation metering non-functional software	TDM-1921 , TDM-1083 , TDM-35 , TDM-1085
TDM-338	Draft of DC architecture spec		Mar 18, 2019	Jeremy Kincaid	DONE	1848 1850		TDM-452 , TDM-341 , TDM-340 , TDM-318 , TDM-339 , TDM-327
TDM-389	Create Functional, Environmental, Certification test plan		Oct 20, 2021	Jeremy Street-Thomas	DONE	1850 1902 1903 1904		TDM-984 , TDM-983 , TDM-452 , TDM-264 , TDM-390
TDM-387	Use existing SW tasks to publish SW architecture specifications		Mar 18, 2019	Jeremy Kincaid	DONE	1850 1902 1905		TDM-855 , TDM-459
TDM-296	Design Random Number Generator		Jul 01, 2019	Jeremy Kincaid	DONE	1850 1902 1903	Design software	TDM-1931 , TDM-385 , TDM-173 , TDM-35
TDM-388	Create specifications, test matrix & coverage status		Oct 20, 2021	Jeremy Street-Thomas	DONE	1905		TDM-452
TDM-453	Publish system requirements draft		Mar 18, 2019	Michael Walton	DONE	1905		TDM-458 , TDM-457 , TDM-455 , TDM-454 , TDM-452
TDM-896	Update planning		Mar 18, 2019	Jeremy Kincaid	DONE	1909		TDM-898 , TDM-897 , TDM-900 , TDM-899 , TDM-459
TDM-905	Certification Questions to VDE		Oct 20, 2021	Jeremy Street-Thomas	DONE	1915		TDM-975 , TDM-974 , TDM-973 , TDM-972 , TDM-971 , TDM-970 , TDM-452 , TDM-979 , TDM-978 , TDM-977 , TDM-976 , TDM-1025 , TDM-1306 , TDM-1307 , TDM-1948 , TDM-1949 , TDM-964 , TDM-963 , TDM-962 , TDM-961 , TDM-960 , TDM-927 , TDM-969 , TDM-924 , TDM-968 , TDM-923 , TDM-967 , TDM-966 , TDM-1318 , TDM-1319 , TDM-951 , TDM-950 , TDM-917 , TDM-916 , TDM-915 , TDM-959 , TDM-958 , TDM-957 , TDM-956 , TDM-954 , TDM-1001 , TDM-1320 , TDM-1321 , TDM-1004 , TDM-1368 , TDM-918 , TDM-982 , TDM-906 , TDM-949 , TDM-948 , TDM-1050 , TDM-947 , TDM-946 , TDM-945 , TDM-1334 , TDM-1016 , TDM-1019
TDM-548	DFMEA mechanical / industrial design scope		Oct 20, 2021	Bernie Walsh	DONE	1915	Main_Focus	TDM-932
TDM-530	DFMEA all electrical scope, including changes to charger bus bars		Oct 20, 2021	Jeremy Street-Thomas	DONE	1915		TDM-527
TDM-477	Publish electrical system specs draft		Oct 20, 2021	Unassigned	DONE	1915		TDM-466
TDM-467	Publish mechanical / industrial system specs draft		Apr 11, 2019	Bernie Walsh	DONE	1915		TDM-464

TDM-1314	Follow up on quoted standards		Oct 20, 2021	Jeremy Street-Thomas	DONE	1915		TDM-452
TDM-1051	[Cancelled] Release memory requirements		Nov 29, 2019	James Sneyd	DONE	1915		
TDM-1043	Release updated requirements		Oct 20, 2021	Jeremy Street-Thomas	DONE	1915		
TDM-1042	Release new PCB allowed dimensions to PCB designers		Nov 12, 2019	Oliver Gruber	DONE	1915		TDM-1041
TDM-1007	Review heat sink requirements for dispenser integration		Aug 12, 2019	Ben Guymer	DONE	1915		TDM-464 , TDM-1008
TDM-990	Update documentation FPGA		Apr 11, 2019	Alex Rudzki	DONE	1915		TDM-452 , TDM-913 , TDM-922
TDM-2198	General code reviews		Oct 20, 2021	James Sneyd	DONE			TDM-2199
TDM-2194	FPGA encryption		Mar 03, 2020	Alex Rudzki	DONE			TDM-2195 , TDM-2196 , TDM-2177
TDM-2193	Backlight display monitoring		Mar 03, 2020	Alex Rudzki	DONE			TDM-2201 , TDM-2200 , TDM-2177
TDM-2190	Automate Interceptor IP configuration		Dec 18, 2019	Alex Rudzki	DONE			TDM-2191 , TDM-2192 , TDM-2177
TDM-2178	Update pixel intercept module for PK		Dec 18, 2019	Alex Rudzki	DONE			TDM-2179 , TDM-2180 , TDM-2182 , TDM-2181 , TDM-2184 , TDM-2185 , TDM-2188 , TDM-2177
TDM-2058	FreeRTOS missing interrupts (ADS1220)		Nov 20, 2019	James Sneyd	DONE			
TDM-2055	Network-Attached Debug Printing		Nov 20, 2019	James Sneyd	DONE			TDM-936 , TDM-1915
TDM-1496	Revise accuracy calculations		Nov 20, 2019	Jeremy Street-Thomas	DONE			TDM-1497 , TDM-938
TDM-401	Draft NPI plan for DC meter project		Oct 20, 2021	Timo Bellgardt	DONE			
TDM-397	Create draft WBS and prepare		Mar 18, 2019	Alex Rudzki	DONE			TDM-452 , TDM-308
TDM-396	Create style guide for the Terminal		Oct 20, 2021	James Sneyd	DONE		software user-interface	TDM-1915
TDM-378	Networking /Comms Stack Integration: Explore Alternative RTOS		Nov 12, 2019	James Sneyd	DONE		Firmware	TDM-1028
TDM-372	Networking /Comms Stack Integration: lwIP Application		Nov 12, 2019	James Sneyd	DONE			TDM-931 , TDM-366 , TDM-377 , TDM-365 , TDM-375 , TDM-374 , TDM-1933 , TDM-373

TDM-366	Networking /Comms Stack Integration: mbedTLS Application		Nov 12, 2019	James Sneyd	DONE	Firmware	TDM-346 , TDM-1028 , TDM-230 , TDM-372 , TDM-371 , TDM-1914
TDM-365	Networking /Comms Stack Integration: FreeRTOS		Nov 12, 2019	James Sneyd	DONE	Firmware	TDM-369 , TDM-368 , TDM-367 , TDM-1028 , TDM-1933 , TDM-372 , TDM-370
TDM-364	Networking /Communicatio n Stack Integration Concept		Nov 12, 2019	James Sneyd	DONE		TDM-931 , TDM-1933
TDM-1466	Documentation M2 FW		Nov 29, 2019	James Sneyd	DONE		TDM-1469
TDM-1425	Document DC Meter software build process		Oct 20, 2021	James Sneyd	DONE	documentat ion process software	
TDM-1365	Design HW Controller		Nov 12, 2019	Richard Kosik	DONE		TDM-938
TDM-1364	Design HW AFE		Nov 20, 2019	Alex Rudzki	DONE		TDM-1464 , TDM-1366 , TDM-1465 , TDM-938 , TDM-1460 , TDM-1462
TDM-1363	Design HW FPGA		Nov 12, 2019	Vivien Wong	DONE		TDM-1367 , TDM-929 , TDM-938
TDM-1357	FPGA task planning		Nov 20, 2019	Alex Rudzki	DONE		TDM-1358 , TDM-452
TDM-1354	Verify FPGA operation		Oct 20, 2021	Alex Rudzki	DONE		TDM-1353 , TDM-1355 , TDM-1356
TDM-1337	Develop FPGA - Firmware		Oct 20, 2021	Jakub Tkaczyk	DONE		TDM-1342 , TDM-986 , TDM-1343 , TDM-61 , TDM-1344 , TDM-1356 , TDM-63 , TDM-62 , TDM-64 , TDM-1341 , TDM-1352
TDM-1336	Develop FPGA - Pixel Intercept module		Apr 17, 2020	Alex Rudzki	DONE		TDM-986 , TDM-2083 , TDM-2095 , TDM-2098 , TDM-1350 , TDM-1340 , TDM-1351 , TDM-2166 , TDM-2132 , TDM-2165 , TDM-2134 , TDM-1355 , TDM-60 , TDM-1840 , TDM-65 , TDM-1338 , TDM-1339
TDM-2075	FPGA interfacing		Nov 21, 2019	Alex Rudzki	DONE		TDM-986 , TDM-2076
TDM-7	Technical research		Mar 18, 2019	Alex Rudzki	DONE		TDM-325 , TDM-321 , TDM-376 , TDM-452 , TDM-361
TDM-354	lwIP Integration (No Operating System)		Jul 01, 2019	James Sneyd	DONE		TDM-359 , TDM-358 , TDM-357 , TDM-334 , TDM-356 , TDM-355 , TDM-1933 , TDM-35
TDM-329	Evaluate CAN bus message fragmentation (No Operating System)		Jul 01, 2019	James Sneyd	DONE		TDM-1908 , TDM-359 , TDM-333 , TDM-332 , TDM-331 , TDM-330 , TDM-35
TDM-310	Update/refine /test the ADC subsystem M1		Nov 25, 2019	James Sneyd	DONE		TDM-931 , TDM-1920 , TDM-24
TDM-187	configuration updates (or parts thereof)		Nov 12, 2019	Andy Meldrum	DONE	DC-Meter-Conf	TDM-1901 , TDM-936 , TDM-1915
TDM-183	DC Meter Run Time Self Test		Jul 01, 2020	Alex Rudzki	DONE	functional software	TDM-148 , TDM-2292 , TDM-1180 , TDM-2294 , TDM-2311 , TDM-2310 , TDM-2313 , TDM-1124 , TDM-2401 , TDM-2312 , TDM-2315 , TDM-2337 , TDM-2314 , TDM-2317 , TDM-76 , TDM-2316 , TDM-2308 , TDM-2319 , TDM-2318 , TDM-2309 , TDM-1904
TDM-169	SW Documentation X1		Mar 18, 2019	Jeremy Kincaid	DONE		TDM-35
TDM-168	Charger Software		Oct 20, 2021	Unassigned	DONE		TDM-1917
TDM-167	Event Log		Nov 12, 2019	Andy Meldrum	DONE		TDM-931 , TDM-1127 , TDM-1129 , TDM-1923 , TDM-1060 , TDM-1173

TDM-166	Test Environment X1		Mar 18, 2019	Jeremy Kincaid	DONE		TDM-35
TDM-165	Menu for technician		Oct 20, 2021	James Sneyd	DONE	SDD UI adaptation software status-reporting	
TDM-164	Displaying meter and charge information		Nov 26, 2019	James Sneyd	DONE		TDM-1908 , TDM-1004
TDM-163	Voltage Supervisor		Jul 12, 2021	James Sneyd	DONE		TDM-1907
TDM-162	User access /permissions scheme		Apr 17, 2020	Alex Rudzki	DONE		TDM-2266 , TDM-1925 , TDM-150 , TDM-190
TDM-161	Design Communications SW		Oct 20, 2021	James Sneyd	DONE		
TDM-159	Circular buffer mechanism X1		Jul 01, 2019	Jeremy Kincaid	DONE		TDM-35 , TDM-1937
TDM-158	Upgrading (Bootloader)		Oct 20, 2021	James Sneyd	DONE		TDM-1145 , TDM-2056
TDM-157	LED toggling (if no display)		Jul 01, 2019	James Sneyd	DONE	systems-engineering	TDM-980
TDM-156	Create a common set of status information for basic diagnostics		Oct 20, 2021	James Sneyd	DONE		TDM-1028 , TDM-1915
TDM-155	Event Log Recording /Reporting		Oct 20, 2021	James Sneyd	DONE	record-report	TDM-1113 , TDM-1126 , TDM-1137 , TDM-1116 , TDM-1924
TDM-154	Usage Session Measurements Recording /Reporting		Oct 20, 2021	James Sneyd	DONE	record-report	TDM-1083 , TDM-1085
TDM-153	UID generation for charge sessions		Oct 20, 2021	James Sneyd	DONE	record-report	TDM-1921 , TDM-1086 , TDM-1087
TDM-152	Cumulative tariff registers summing all charge sessions		Jul 01, 2019	Andy Meldrum	DONE	record-report	TDM-1166 , TDM-1921 , TDM-1945 , TDM-35 , TDM-1163 , TDM-1164
TDM-151	Tariff settings		Jul 01, 2019	Andy Meldrum	DONE	SDD tariff	TDM-1166 , TDM-561 , TDM-1945 , TDM-35 , TDM-1165
TDM-150	Security Keys Management		Oct 20, 2021	James Sneyd	DONE		TDM-162 , TDM-1929
TDM-149	Develop SW for setting and retrival of DC Meter Configuration		Oct 20, 2021	James Sneyd	DONE	SDD adaptation functional software	TDM-1901
TDM-148	DC Meter Power On Self Test		Oct 20, 2021	Alex Rudzki	DONE	software	TDM-1124 , TDM-2315 , TDM-2337 , TDM-76 , TDM-2316 , TDM-2319 , TDM-2292 , TDM-2318 , TDM-2294 , TDM-1903 , TDM-183 , TDM-2320
TDM-147	De-Risking activities		Mar 29, 2019	Jeremy Kincaid	DONE		TDM-35 , TDM-171 , TDM-170
TDM-92	X1 Design verification HW display interceptor		Apr 18, 2019	Alex Rudzki	DONE		TDM-998
TDM-81	Backend integration on controller		Oct 20, 2021	James Sneyd	DONE		TDM-1915

TDM-80	NTP packet forwarding for ethernet over CAN		Apr 26, 2019	James Sneyd	DONE		
TDM-72	Revisit AFE ovenisation control		Nov 08, 2019	Alex Rudzki	DONE		TDM-980
TDM-34	X1 Design verification HW controller		Apr 18, 2019	Alex Rudzki	DONE		TDM-998
TDM-33	X1 Design verification HW high current sensor		Apr 18, 2019	Alex Rudzki	DONE		TDM-998
TDM-32	Design voltage sense point enclosures		Nov 12, 2019	Bernie Walsh	DONE	Input_required	TDM-932
TDM-31	Design meter main enclosure		Mar 18, 2019	Bernie Walsh	DONE		TDM-489 , TDM-488 , TDM-487 , TDM-486 , TDM-29
TDM-30	Shunt resistor PCB mounting bracket		Apr 10, 2019	Bernie Walsh	DONE		TDM-29
TDM-27	Develop FPGA - CPU and Comms module		Nov 26, 2019	Jakub Tkaczyk	DONE		TDM-986 , TDM-353 , TDM-66 , TDM-2131 , TDM-59 , TDM-1350 , TDM-2133 , TDM-1344 , TDM-1355 , TDM-1345 , TDM-1346 , TDM-1347 , TDM-1348 , TDM-1349 , TDM-2109
TDM-26	Design FW for controller X1		Apr 10, 2019	Jeremy Kincaid	DONE		TDM-35
TDM-24	Design Controller		Apr 01, 2019	Jeremy Street-Thomas	DONE		TDM-315 , TDM-314 , TDM-303 , TDM-940 , TDM-311 , TDM-322 , TDM-310 , TDM-305
TDM-23	Design HW AFE		Mar 18, 2019	Alex Rudzki	DONE		TDM-50 , TDM-907 , TDM-44 , TDM-904 , TDM-49 , TDM-305
TDM-294	X1 Design HW FPGA		Mar 29, 2019	Vivien Wong	DONE		TDM-326 , TDM-348 , TDM-347 , TDM-985 , TDM-324 , TDM-323 , TDM-344 , TDM-342 , TDM-56 , TDM-284 , TDM-283 , TDM-272
TDM-272	Bring up current TRI105 HMI on Apalis module with FPGA overlay		Oct 20, 2021	Jakub Tkaczyk	DONE		TDM-2059 , TDM-286 , TDM-294
TDM-882	Working With Code Composer Studio		Mar 18, 2019	Andy Meldrum	DONE		TDM-459
TDM-862	Decimal Fixed Point Numbers		Jul 10, 2019	Andy Meldrum	DONE		TDM-1079 , TDM-1921 , TDM-35 , TDM-1085
TDM-854	Publish SW requirements		Mar 18, 2019	Jeremy Kincaid	DONE		TDM-560 , TDM-559 , TDM-459
TDM-607	Upload final BOM		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-606	Preliminary process capability study plan		Oct 20, 2021	Dion Schulz	DONE		TDM-562
TDM-605	Measurement system analysis plan		Oct 20, 2021	Dion Schulz	DONE		TDM-562
TDM-604	Process sheets		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-603	Pre-launch control plan		Oct 20, 2021	Dion Schulz	DONE		TDM-562
TDM-598	PFMEA		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-593	Characteristics matrix		Oct 20, 2021	Harry Watson	DONE		TDM-562

TDM-588	Floor plan layout		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-583	Process flow chart		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-578	Product / process qualiy system review		Oct 20, 2021	Dion Schulz	DONE		TDM-562
TDM-573	Packaging standards and specifications		Oct 20, 2021	Harry Watson	DONE		TDM-562
TDM-568	Develop DFM strategy		Oct 20, 2021	Timo Bellgardt	DONE		TDM-562
TDM-566	Develop manufacturing plan / design production line		Oct 20, 2021	Timo Bellgardt	DONE		TDM-567 , TDM-562
TDM-563	Develop Quality plan		Oct 20, 2021	Dion Schulz	DONE		TDM-565 , TDM-452
TDM-529	Design busbar changes		Mar 18, 2019	Bernie Walsh	DONE		TDM-535 , TDM-538 , TDM-537 , TDM-29 , TDM-536
TDM-528	Design harnesses		Oct 20, 2021	Cameron Swales	DONE		TDM-934
TDM-472	Publish FW / HW system specs draft		Nov 11, 2019	Jeremy Street-Thomas	DONE		TDM-476 , TDM-465 , TDM-475 , TDM-474 , TDM-473
TDM-450	Configurable language pack for setting and retrival of DC meter Configuration		Oct 20, 2021	James Sneyd	DONE	SDD UI adaptation	TDM-1901 , TDM-449
TDM-1322	Design process for handling of cryptographic keys		Oct 20, 2021	Jeremy Street-Thomas	DONE		TDM-562
TDM-1316	IONITY		Oct 20, 2021	Unassigned	DONE		TDM-1315 , TDM-1317
TDM-1052	[Cancelled] Review cooling options		Nov 29, 2019	Michael Walton	DONE		TDM-932
TDM-1049	Update specifications: need to monitor shunt temperature		Nov 12, 2019	Jeremy Street-Thomas	DONE		TDM-932
TDM-1048	Report ISD DC meter shunt cooling options to Bernie		Nov 12, 2019	Bernie Walsh	DONE		TDM-932
TDM-1035	Present cooling concept		Nov 12, 2019	Bernie Walsh	DONE		TDM-932
TDM-1031	Redesign meter enclosure		Nov 12, 2019	Bernie Walsh	DONE	Main_Focus	TDM-932
TDM-1024	Validate aliasing assumptions (for sigal sampling)		Oct 20, 2021	James Sneyd	DONE		TDM-1009
TDM-1023	Trial "Enterprise Architect" SW		Nov 08, 2019	Jeremy Kincaid	DONE		

TDM-1021	Do we need to be ready for dual cable charging pin voltage sensing?		May 07, 2019	Aidan Clarke	DONE	TDM-1020
TDM-1017	Component-level validation of TRI110-04v1 FPGA Interceptor PCB X1		Nov 20, 2019	Vivien Wong	DONE	TDM-1018 , TDM-998
TDM-1014	Fail to select appropriate RTOS		Nov 20, 2019	James Sneyd	DONE	TDM-1015 , TDM-1009
TDM-1011	FPIO ethernet comms is disturbed by electrical noise		Nov 14, 2019	Nicholas Keeling	DONE	TDM-1013 , TDM-1009
TDM-1010	CAN loading impact on meter / display / backend communcation		May 28, 2019	Dion Sumpton	DONE	TDM-1012 , TDM-1009
TDM-999	Component-level validation of TRI110-03v1 FPGA carrier PCB X1		Nov 20, 2019	Vivien Wong	DONE	TDM-1000 , TDM-998
TDM-989	Test Environment M1		Oct 20, 2021	James Sneyd	DONE	
TDM-988	SW Documentation M1		Nov 12, 2019	James Sneyd	DONE	TDM-931 , TDM-1467 , TDM-1468
TDM-981	Replace 5x gland between front and back by bulkhead connector		Oct 20, 2021	Cameron Swales	DONE	
TDM-953	Identify second source for shunt		Oct 20, 2021	Jeremy Street-Thomas	DONE	TDM-934
TDM-952	Redesign shunt mounting		Nov 12, 2019	Bernie Walsh	DONE	TDM-932
TDM-943	Design and Implement the ADC subsytem X1		Jul 01, 2019	Jeremy Kincaid	DONE	TDM-1920 , TDM-926 , TDM-35 , TDM-944 , TDM-965
TDM-942	Circular buffer mechanism M1		Oct 20, 2021	James Sneyd	DONE	
TDM-941	Design FW Controller M1		Oct 20, 2021	James Sneyd	DONE	
TDM-930	M1 Design HW FPGA		Nov 12, 2019	Vivien Wong	DONE	TDM-929
TDM-928	Design Remote Display		Nov 26, 2019	Vivien Wong	DONE	TDM-2059
TDM-920	Update documentation Meter		Mar 25, 2019	Alex Rudzki	DONE	TDM-452 , TDM-921

135 issues

List of key documents and references

Please use google to translate German material: https://translate.google.com/?client=firefox-b&um=1&ie=UTF-8&hl=en&client=tw-ob#view=home&op=translate&sl=de&tl=en

document / reference	comment	link
statement of work	written by Martin	Statement of Work - DC meter - 190219 - V1.2.docx
system requirements	written by Michael	TRI110.026v2 DC Meter system spec.docx
architecture specification	written by Deepak	
architecture specification	written by Jeremy	https://wiki.tritium.com.au/x/Owx9AQ
VDE draft standard	extracted from online tool and translated by google we have an original paper hardcopy in German	VDE AR E 2418 3 100 draft E.docx
WBS and requirements	in Jira	https://jira.tritium.com.au/projects/TDM?selectedItem=jp.ricksoft.plugins.wbsgantt-for-jira:wbsgantt-project
design folder	important subfolders: 001 engineering reference 003 block diagram 011 LCD screen integration 013 SAFE initative (transparency SW) 022 FPGA selection 024 Standards Compliance, esp. requirements matrix	Z:\Tritium\TRI110 - DC Meter\00 - Documents
description of "convenient solution"	This is an unofficially recommended solution (coming from PTB) Documents are in German and have been translated by google	E2E-Sicherheit early release rev 1 TRANSLATION.pdf.docx read me info4 rev 1 TRANSLATION.pdf.docx Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.025 Convenient Solution - info from Angelika and PTB 190123
Has2Be	Download beta version transparency SW (German) General info (German)	https://has-to-be.com/de/development-resources/transparenzsoftware/beta-download-transparenzsoftware/ https://has-to-be.com/de/development-resources/transparenzsoftware/
Testing labs	CSA (close to Munich) VDE (close to Frankfurt)	https://www.csagroup-bayern.org/ https://www.vde.com/tic-de
article about public key infrastructure	about Germany / in German	https://www.electrive.net/2019/01/12/bundesnetzagentur-erweitert-ladesaeulenkarte/
article about extension of deadline	about Germany / in German	https://www.electrive.net/2018/12/17/eichrecht-bei-ladeinfrastruktur-gemessener-gleichstrom/

M1 Prototype Hardware Issues

Design issues identified to be corrected:

#	Item	Issue	Remedy	Status	Completed
1	Controller PCBA	U9 (11345) has no opening on top paste.	Update footprint from library	Fixed in library. Part no longer on board	•
2	Controller PCBA	U9 (11345) is planned end of life	Replace with TC58CVG2S0HRAIG		•
3	Controller PCBA	Several single line items that don't seem critical	Reduce single line items where possible		•
4	Controller PCBA	Ringing on mosfet gates	Change DNF connecting mosfet driver ground from GND to GND_PWR (remove R82/R89 and fit R83/90)		•
5	Controller PCBA	Potential for mosfet gates to float	Add 10k pull down to mosfet gate		•
6	Controller PCBA	Fans/thermistor connector is SMD and won't be able to be picked up by ATE	Place test points on fans /thermistor lines		•
7	Controller PCBA	Fan feedback has minimum voltage of 0.3V. Inline resistor causes voltage divider.	Move pull up to connector side.		•
8	Controller PCBA	Fan feedback is being pulled up to above 3.3V when off. Fan power is trying to return through feedback.	Add optocoupler for speed using fan power		•
9	Controller PCBA	Requirement may have changed to need more memory	Add second flash part on bottom side (default DNF)		•
10	Controller PCBA	Will TPM be used?	Default is fitted.	Should be able to be removed later if not used.	•
11	Controller PCBA	External reference for ADC doesn't really provide value	Change to DNF, use 3.3V reference and change temp sensor bias resistors to 1%	Firmware must change to use 3.3V	•
12	Controller PCBA	Lower ovenisation power than expected. Track resistance is higher than expected and ovenisation voltage under load is lower.	Change transformer to 1:1.75		•
13	Controller PCBA	Cutouts for baffle have uncertain radius	Place known radius for cutout		•
14	Controller PCBA	New injected enclosure would interfere with corners of the board	Increase corner radius on board corners to avoid interference	Confirm to what size	•
15	Controller PCBA	Controller requires conformal coating for creepage	See if creepage (10mm) can be obtained to avoid this		•
16	AFE PCBA	Heat retention around AFE PCBA. Put foam inside the canister area, over the can, or over the whole board. Foam material? Silicon conformal coating?	Metal can by itself is sufficient for our needs	No action needed	•
17	AFE PCBA	Q1 (5528) footprint is quite wide	Change footprint		•
18	AFE PCBA	HV Detect circuitry can have voltage applied on other cable	Switch high voltage lines to allow two high accuracy voltage channels using ASSR-601JV-000E	Needs review	•

19	AFE PCBA	One signal across the barrier is currently unused	Now used for one wire interface		•
20	AFE PCBA	Could eliminate production needing to place formex over RTC cap	Increase cutout/copper-free distance around capacitor		•
21	AFE PCBA	Connection of temperature sensor to shunt needs to be reinforced isolation and well thermally coupled	Use one wire interface and place on AFE board.		•
22	AFE PCBA	Input power filters aren't used	Remove beads from board		•
23	Enclosure	Labelling panel isn't ideal for production solution	Design metal riveted plate on outside for required info	Confirm required information needed to be displayed	•
24	Enclosure	Inserts for board mounting need more material around them	Require more space around AFE mounting points	Need to support with minor AFE board change too.	•
25	Enclosure	Possibility for Fan wires to be pushed onto heatsink	Extend a cover over wires on the fan		•
26	FPGA Interceptor PCBA (v2)	Level shifter (TXB0106) should be replaced with an open-drain/push-pull level shifter (TXS0108) for TPM chip so that pull-up resistors can be used with it. The current level shifter (TXB0106) works for all signals except the open-drain pin PIRQ# on the TPM chip.	Unfit R63 and R65. Pin is driven by FPGA Fit R64 with 100ohm so that PIRQ# is held high. Possibly put in a buffer on PIRQ# between TPM and level shifter	Used remedy as of now.	•
27	FPGA Interceptor PCBA (v2)	EXT_CLK should be connected to "P-side" of clock capable diff pair input, (ref UG472 : page 49)	TPM_CLK_FPGA should be able to be moved to pin 'P4' as it probably doesnt need a dedicated clock pin	Presently implemented a workaround in FPGA fabric, but is sub-optimal and not recommended	•
28	FPGA Interceptor PCBA (v2)	FPGA part should be maintained as -2 speed grade (similar to v1)- CID 10867. On the V2, due to stock issue, -1 speed grade was used as a substitute CID 12658	Use CID 10867 -2 speed grade FPGA part for future builds.		•
29	FPGA Interceptor PCBA (v2)	Insufficient RAM for video buffer and microblaze CPU	Change to FPGA part with larger RAM with same package: XC7S50-2FTGB196I		•

M2 Prototype Hardware Issues

Design issues identified to be corrected:

#	Item	Issue	Remedy	Status	Completed
1	Controller PCBA	J3 has + and - swapped around from Tritium standard. -ve on top is standard	Should swap these pins around and also consider which location a 2 pin minifit would fit to allow harness re-use	Optional: Decided not needed change	•
2	Controller PCBA	Uncertainty if IR LED pulse energy is strong enough	Potentially decrease resistors (2 parallel x 390R used for testing works with less than ideal IR receiver)	R69 and R72 3164 on v3a	•
3	Controller PCBA	(Cosmetic) Note on UVLO has wrong comment on VREF setting	Change comment to VREF=0	To be done on next update	•
4	Controller PCBA	(Cosmetic) Note on fan speed low pass filter assumes only 1 pulse per revolution	Change comment to 10krpm = 333.33hz = 3ms (2 pulses per revolution)	To be done on next update	•
5	Controller PCBA	(Cosmetic) TP30 "TEMP_SNS" is missing overlay for TP name	Add name to overlay	To be done on next update	•
6	Controller PCBA	EMC Iso power noise	Fit Y caps; Change Y cap connection to GND_PWR and ISO_GND-prebeads (closer to transformer) Change Y caps to 2x CID:11147		•
7	Controller PCBA	EMC Iso power noise	Fit two 100pF caps on iso secondary near transformer (parallel C67/C68) (CID:143)		•
8	Controller PCBA	EMC Iso power noise	Change drive resistors to 15R (R71/R74) (CID: 2681)		•
9	Controller PCBA	EMC Iso power noise	Change snubber resistors to 4.7R (R73/R26) (CID: 101)		•
10	Controller PCBA	EMC noise	Fit 100nF caps to earth (C73/C75)		•
11	Controller PCBA	EMC ethernet noise	Fit 1nF to earth for ethernet termination (replace DNF R1) (CID: 7905)		•
12	Controller PCBA	EMC ethernet noise	Fit U4 ESD protection		•
13	Controller PCBA	EMC fixes may require addition space on board	Can remove the secondary DNF iso transformer circuitry		•
14	Controller PCBA	Compliance - CTI not specified	Specific that CTI >=175 in the manufacturing details		•
15	Controller PCBA	Compliance - creepage; for slot to extend creepage must be >1mm	Increase slot to 1.2mm		•
16	AFE PCBA	Q2-4 Fets have incorrect part number on schematic	Update comment	(documentation only, no physical change)	•
17	AFE PCBA	Both opto pairs turn on at once. Leakage through FET gate	Change R40: 100k pull down to 10Meg / Change to flip /flop;comparator?	R40 DNF (1259) on v3	•

18	AFE PCBA	Temperature sensor doesn't produce pulses.	Change pull up resistor to 10k and mosfet to npn transistor Possible future revision: change to comparator	R16 360 on v3 Q4 12540 on v3 This is reference design so no comparator needed.	•
19	AFE PCBA	Temperature sensor doesn't produce pulses.	Correct sensor footprint.	TODO on next board revision	•
20	AFE PCBA	Compliance - CTI not specified	Specific that CTI >=175 in the manufacturing details		•
21	DCM Assembly	EMC noise	Add earth cable to power harnesses		•
22	DCM Assembly	EMC noise	Change ethernet cable to shielded one - properly terminated		•
23	DCM Assembly	EMC noise	Add f19 (13184) to ethernet cable near DC meter end		•
24	DCM Assembly	EMC noise	Change RJ45 connector to one that connects the shield (17191)		•
25	DCM ISD HMI Assembly	EMC noise	Add f19 (13184) to ethernet cable near HMI end		•
26	DCM ISD HMI Assembly	EMC noise	Correctly terminate ethernet shield to connector		•
27	DCM ISD HMI Assembly	EMC noise	Spray paint HMI's plastic back cover internally with EMI shielding paint (CID: 5154)		•
28	DCM ISD HMI Assembly	EMC noise	Include f19 ferrites on 24V to HMI that already exist in total charger build	No design action needed, already exists	•
29	DCM ISD HMI Assembly	EMC immunity	Add f19 (17022) on 12V to DC meter		•
30	DCM ISD HMI Assembly	EMC immunity	Add f19 (17022) on 24V with a turn to HMI		•
31	DCM ISD HMI Assembly	EMC noise	Add earth braid from bottom right HMI bracket direct to ground point		•
32	DCM ISD HMI Assembly	EMC noise	Add jointing compound on HMI bracket feet		•
33	DCM PK HMI Assembly	EMC noise	Include 2xf19 ferrites on 24V rail that already exist in total charger build	No design action needed, already exists	•
34	DCM PK HMI Assembly	EMC noise	Place f19 ferrite (17022) on 24V and ethernet together close to HMI		•
35	DCM PK HMI Assembly	EMC noise	Connect braid directly to HMI earth stud		•
36	DCM PK HMI Assembly	EMC noise	Place f19 ferrite (17022) on ethernet from HMI with a turn		•

Mechanical placement requirements

The DC meter is comprised of two major mechanical components

1. The main metering module - This module is installed in the output power stage of the charger and measures the output current and voltage
2. Front display intercept module - This module is installed in the HMI to securely display information on the screen

The ambient temperature around the meter components must be within the range of **-40°C to +85°C**

Main metering module

The main metering module has a number of specific installation requirements to function correctly.

- All charger output current must flow through the Shunt (shunt resistor)
- The Shunt shall be installed in the negative output
- **The Shunt must not be connected close to any large heat source (Output fuse or diodes)**
    - Any thermal difference across the Shunt will cause a substantial error in measurement (This will likely be why the ISA meter had such long bus bars internally)
- **The Shunt will generate ~13W of heat when the charger output current is at 500A**
    - It is expected that the shunt will operate at a maximum temperature of 120°C
    - Adequate cooling of the shunt must be provided. The copper bus-bars will likely not be enough
    - Cooling solutions could be
        - Heat sinks to surrounding air, attached to the copper bus-bars
        - Water cooling block
    - Adequate air flow shall be provided around the DC meter
- At lower currents such as in the RT, the shunt will generate ~0.8W at 125A
- The module must measure charger output voltage. Only a positive measurement point is required
- The output voltage is measured by connecting a single wire to an external positive bolt terminal in the charger
- The positive voltage measurement sense point must be after any output diodes (Essentially needs to be as close to the cable connection point as possible)
- The module must have a tamper evident enclosure which encases all of its electronics (Not sure if the Shunt terminals need to be enclosed too)
- The Shunt bolt terminals and positive voltage measurement point will need to have some form of tamper evident seal

Refer to this datasheet for Shunt mechanical dimensions:

"Z:\Tritium\TRI110 - DC Meter\00 - Documents\Research\Technical\front end measurement\Shunt\Vishay\WSBS8536-40 29-May-2018.pdf"

Refer to this document for conceptual PCB layout. At present use the Shunt dimensions as a reference for size:

"Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.021 PK High Current Meter mechanical location\TRI110.021 DRAFT circuit board layout.pdf"

Refer to this document for the general thinking of the meter location in the PK back electronics enclosure:

"Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.021 PK High Current Meter mechanical location\Tritium DC meter location in PK.pdf"

Front display intercept module

- This must be installed between the HMI computer and LCD screen (For PK)
- The module is just a PCB. It will need some tamper evident seal

No solution present for RT. Possibly have a small LCD screen similar for ChargePoint HMI?

Refer to this PCB that is included in the Ionity HMI for mechanical dimensions:

"Z:\Tritium\TRI110 - DC Meter\05 - FPGA Carrier Dummy\Hardware v1\Project Outputs for TRI110-05-v01\TRI110-05-v01 - PCB MANUFACTURING DETAILS.PDF"

There is another PCB that will be on top of this one, it might add ~20mm of height.

Software Development Plan

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  - Table - List of easily reusable modules for other products/projects
- Table - List of sources for the Product Requirements
- Table - WELMEC 7.2 Risk Class Definition
- Table - WELMEC 7.2 Risk Assessment

Scope

Identification

This document contains details about the software development process for the DC Meter project identified by the Tritium internal project code: TRI110. In the Tritium issue management system known as JIRA, this project has the code: TDM and can be found at https://jira.tritium.com.au/projects/TDM/summary. In Tritium's Wiki, documentation relating to this project can be found at https://wiki.tritium.com.au/display/RND/DC+Meter.

System Overview

The need for the DC Meter is a consequence of the German authorities making it mandatory for all Electric Vehicle (EV) Charging Stations to contain a certified DC Meter suitable for billing purposes as specified in the standard document EN50470-1/-3. The values measured/calculated by the DC Meter are used by government authorities and end users to verify billing information from the Charge Point Operators (CPO) managing the EV Charging Stations.

The DC Meter is analogous to an AC Electricity Smart Meter but for EV Charging stations and instead of measuring the total AC usage to the EV Charging station, it measures the DC usage for charging the vehicle and specifically excludes power to run the EV Charging station, power lost in cabling and other losses.

Functionality includes (but not limited to):

1. accurately measure the DC usage when charging a vehicle
2. calculate the energy delivered to the user
3. keep track of charge session duration and time started
4. report the measured and calculated values to
    a. a display to the user and
    b. an independent server
5. be cryptographically secure in storage and transmission of information.

There are existing Tritium EV Charging stations installed at various site in Germany that will need to be retrofitted with the DC Meter Solution. As a consequence there are internal Tritium requirements that impact the DC Meter with regards to physical size and shape as it needs to fit in existing systems of different models and new systems yet to be built.

German installation sites include:

- Brohtal Ost
- Brohtal West
- Himmelkron
- Gruibingen Süd
- Ringdalskogen

- Hohenwarsleben
- Wunnenstein Ost
- Salzbergen
- Kjerlingland

- Wunnenstein West
- Mahlberg West
- Eichenzell
- Hohenlohe Süd

Different models of EV Charging Stations have different physical sizes and different sized Display Panels. The DC Meter also needs to fit inside these EV models and use their screens (as the DC Metering information is superimposed onto the existing Display Panel instead of having it's own screen).

System block diagram

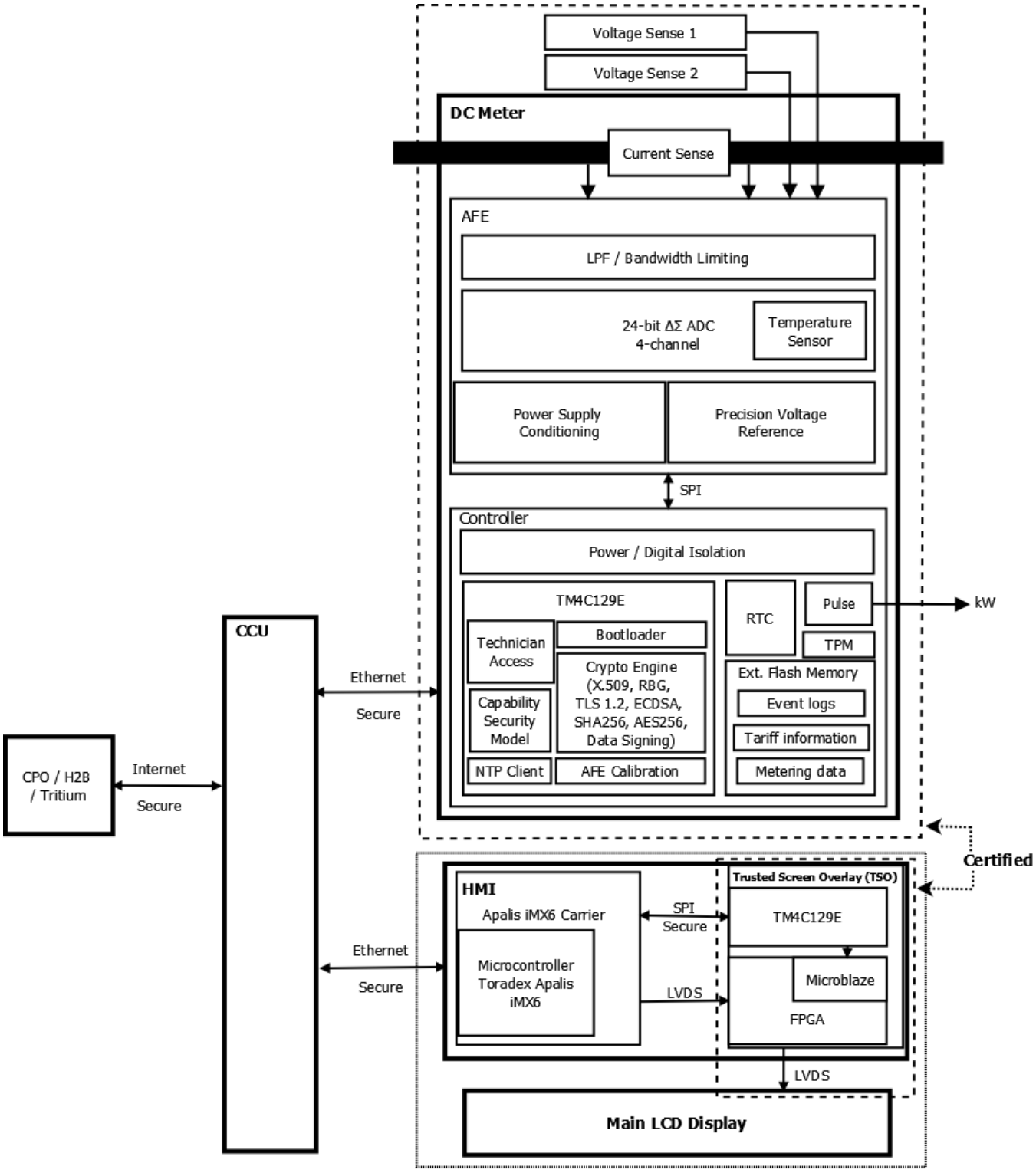


Figure – Block diagram of the DC Meter

Document Overview

As part of the certification requirements of the DC Meter, we need to show we followed a development plan, which outlines processes, tools, and deliverables as part of the development process. This document is commercially sensitive and shall be considered "Commercially in Confidence".

Relationship to Other Plans

Table - Relationship to Other Plans

Document	Subordinate/Equal/Superior
Project Management Plan	Subordinate
Test Plan	Superior
Certification Plan	Equal
Installation/Support Plan	Equal
DC Meter System Specification	Subordinate
DC Meter Firmware Design	Superior

Referenced Documents

Table - Referenced Documents

ID	Reference
R1	**VDE AR E 2418 3 100 draft E** Latest draft (as of 16 Oct 2018) of the system requirements from the German authorities (PTB) used for requirements on the DC Meter.
R2	**WELMEC Guide 7.2** Technical guidance for the application of the Measuring Instruments Directive (MID) for software-equipped measuring instruments.
R3	**WELMEC Guide 7.1** Development of Software Requirements
R4	**POL.003 Tritium C Coding Style & Guidelines** Document containing layout of C code, guides to naming convention

Overview of Required Work

Requirements as specified in the VDE[R1] are to be reviewed and used to create a software work breakdown structure. References to other documents will be reviewed for secondary requirements imposed on the system.

Documentation that needs to be submitted as part of the certification process needs to be identified and listed. Tritium have templates that will be used when providing these documents externally. All documents will need to be reviewed and approved before being released. All documents will initially be created in Confluence and later exported using Tritium Word Templates. Any released document will also be saved in Confluence with the project and optionally stored by Tritium's internal Quality and CI department.

Tests need to be performed on the software to verify the quality and reports are used as part of the deliverables for certification.

The DC Meter project is currently fully funded by Tritium and worked on by Tritium employees. The table below lists the staff working on it:

Table - DC Meter Team

Name	Position	Responsibilities

Aidan Clarke	Product Manager	
Nicholas Keeling	Power Electronics Module Lead	
Alex Rudzki	Hardware/Firmware Lead	AFE sensor board and TSO hardware
Jeremy Kincaid	Software Engineer	All firmware/software for the Controller board and remote LCD
Vivien Wong	Hardware Engineer	FPGA design
Andy Meldrum	Contract Software Developer	All firmware/software for the Controller board and remote LCD
James Sneyd	Firmware Engineer	All firmware for the Controller board and remote LCD
Jeremy Street-Thomas	Electronics Engineer	Hardware for the TM4C Controller board
John Huynh	Software Developer	Software for Trusted Technician Interface and Automated Testing
Hai Nuygen	Software/Electronics Developer	Automated Testing
Jakub Tkaczyk	Module Lead HMI	Trusted Screen Overlay TM4C, Microblaze, communications to DC Meter

Plans for Performing General Software Development Activities

Software Development Process

The software development process will follow the repetitive/iterative cycle as listed below:

1. Requirements Analysis
   Although this step is seen as the initial step, requirements are re-evaluated at regular intervals to catch anything that may have been missed the first time, or if the initial response to the requirement(s) were the best option, or if new information has been learned that leads to a different approach to meeting the requirement. This step should lead to a Work Breakdown Structure (WBS) identifying sub-tasks and level of effort with mapping to original requirements.
2. De-Risking Activities
   If there are requirements that are difficult to estimate the level of effort, or are considered high risk to the success of the project, then a dedicated activity is create to reduce the risk. This activity may be a proof of concept, or an investigation in to different technologies. The end result may be more requirements or a more solid design with estimations. These activities are optional.
3. Development Environment Setup
   Knowing what tools to use, and integrating them together to streamline the process. There may be an initial setup but as the development progresses new tools may be added and old tools replaced or tweaked.
4. Documentation
   The various documents relating to the project are not finalised until the project completes. There may be various versions released throughout but all documents should be considered "living" documents that change as new information comes to light. Documentation may be formal and released to 3rd parties, or may be informal such as JIRA tickets or Confluence/Wiki pages, or even loose documents in the project folder on the server. Formal documents are listed in later in this document, however it is up to the project members to add any other documents they see fit, especially if it help understand the system or explains the reasoning for direction of any part of the solution.
5. Implementation
   This is a combination of software implementation, testing, version control. Items are addressed based on the WBS.
6. Formal Testing
   Official results used in the certification process, peer reviews, also considered internal version of acceptance/functional testing. This step may not occur as often as the other steps as this requires more of the product to be completed before these are practicable.
7. Managing merging to the release branch
   The branch 'master' is the release branch. It should always reflect a working copy of the solution and updates to this branch should be carefully managed to ensure only wanted functionality is added, and the code is bug free.

Build/release cycles will be based on the Kanban methodology. The following releases are envisioned:

1. Prototype 1 - May 2019, using actual hardware but software not yet ready for certification although it should have near full functionality
2. Certification Candidate - Sept 2019, in testing to prove software is ready for certification.
3. Certification Release - Dec 2019, released to the PTB for certification

General Plans for Software Development

Software Development Tool-chain

Software/Firmware developers have the option of developing either in Windows or on a Linux VM running Ubuntu.

Table - Software Tools

Product	Version	Supplier	Notes

TI Code Composer Studio	Windows: 8.2.0.00007 Linux: 8.3.1.00004	http://processors.wiki.ti.com/index.php/Download_CCS#Code_Composer_Studio_Version_8_Downloads	IDE for authoring software/firmware, graphical debugging. This is the TI supported IDE.
TI ARM CGT	18.12.2.LTS	http://www.ti.com/tool/arm-cgt	The compiler created by TI to support the processor. The IDE uses this compiler in its configuration.
TM4C129E Development Board	A	http://www.ti.com/tool/EK-TM4C129EXL	The development kit used to test code on target until the DC Meter hardware is available.
TM4C129ENCPDT	Die Revision A2 Part Revision 3	http://www.ti.com/product/TM4C129ENCPDT?keyMatch=TM4C129ENCPDT&tisearch=Search-EN-everything&usecase=part-number	The core microprocessor on the DC Meter
Git	2.11.1 (Windows) 2.17.1 (Linux)	S:\GIT\Git-2.11.1-64-bit.exe https://git-scm.com/download/win - For Windows for latest built into Ubuntu (sudo apt-get update git)	Code versioning management tool.
TortoiseGit	2.4	S:\GIT\TortoiseGit-2.4.0.0-64bit.msi	The GUI frontend for Git to be used on Windows
GitKraken	6.0.1	http://gitkraken.com	The GUI frontend for Git to be used on Linux
Doxygen	1.8.13 (Linux)	http://www.doxygen.nl/download.html	Used for extracting source comments from code.
Ceedling	Ceedling:: 0.28.3 CMock:: 2.4.5 Unity:: 2.4.2 CException:: 1.3.1	http://www.throwtheswitch.org/ceedling	Used as the test harness. This is compiled into the project (as opposed to being a separate executable) so it is also listed below under reusable libraries. However, to run Ceedling you must install Ruby. Follow the instructions at the link provided. Ceedling includes Unity, CMock, and CException.
Putty		Tritium software drive	Used as the serial terminal to the development kit
PuttyGen		Tritium software drive	to create the key to access Git.
TRI70/00 - Git Setup	ssh://git@git.tritium.com.au/TRI70	Git respository setup scripts for Windows and Linux	Contains some setup scripts to more easily use Git and configures TortoiseGit

Using Code Composer Studio

Useful references: https://software-dl.ti.com/ccs/esd/documents/ccs_portable-projects.html

Code Composer Studio (CCS) is an IDE provided by TI for development of applications based on TI processors.

We are using a TM4C129ENCPDT device

CCS is provided in 32 and 64 bit versions. The 32 bit version is being deprecated. We are using the 64 bit version.

It provides several tool chains, a proprietary toolchain sourced from ARM and a GCC tool chain also sourced from ARM.

CCS is based on the Eclipse Project with TI customisation adding extra functionality for developing for TI microprocessors.

Java

CCS (and eclipse) are Java based, and require a Java Runtime Environment (JRE).CCS 64 bit requires the 64 bit version of Java.

Experience suggests that you are better off using a current version of Java 8. It seems to be Java 9 seems slower.

Project Integration

CCS supports a couple of flavor of projects

CCS Projects - using TI tool chain .. tightly integrated into the IDE .. Build, Debug all work as expected.

C/C++ Projects .. typically using gcc toolchain .. not to tightly integrated into the IDE .. in particular the debug process is not well integrated.

To get the most out of the IDE prefer CCS projects.

Workspace Configuration

Summary

Preferences\Show Advanced Settings
Preferences\ General\Always Run in background - uncheck
Preferences\General\Startup And Shutdown\RSE UI - uncheck
Preferences\General\Show Heap - check.. shown at bottom of screen
Preferences\ C/C++\Build\Console Uncheck Wrap Lines on the Console, Limit Console Output 20000, Delay Console Updates 100
Preferences\General\Workspace\Save Automatically before build
Preferences\General\Editors\Text Editors\Show Line Numbers
Preferences\C/C++/Code Style /Formatter - Import Tritium C Code Template Tritium_C.xml found in Repo/code_composer/templates - Set Tritium C as active profile
Preferences\C/C++/Code Style/Code Templates/Import Tritium Code Templates TritiumCodeTemplates-DoxygenComments.xml
Preferences\C/C++/Editor/Documentation Tool Comments - Doxygen
Preferences\General\Editors\Text Editors\Spelling\ Platform Dictionary - English (UK), User Defined Dictionary to your choosing

CCS (just like Eclipse) encourages the use of multiple work spaces. Work spaces are your friend. The less projects you have in a workspace the easier it will feel to navigate them.. and the less files CCS have to index.

Typically work spaces are not stored as part of the project repo .. they are maintained locally.

Treat work spaces as disposable items .. don't get attached to them. Most of the time this works out fine .. apart from debug configs..

Debug Configs are workspace specific .. but it is possible to import and export them. The real pain is that there a lot of settings in debug configs that have to be just right .. so the trick is to get one working.. then save the debug config somewhere in the project repo.

Remote System Explorer (RSE) - Disable It once per new workspace

Remote System Explorer is built into CCS. The idea is it lets you browse remote systems. The problem is that it tries to do this while you are compiling. It will hang the compile. There are better ways to remotely view files ..

Disable it by unchecking the Windows\Preferences\General\Startup And Shutdown\RSE UI checkbox.

This needs to be done once per Workspace. As far as I know there is no template to make this easier.

Upgrading the USB200 JTAG EMULATOR probe

For instructions refer to the site: http://dev.ti.com/tirex/explore/node?node=AADzJ8Y-La4f7Bi5Ga0TcA__FUz-xrs__LATEST

Compilation Issues

CCS seems to have a myriad of ways of having a compilation fail. Below are some tips to prevent it but another good resource is: http://software-dl.ti.com/ccs/esd/documents/sdto_ccs_build-errors.html#ccs-build-console

Console - uncheck wrap lines

This maybe a root cause of a lot of the compilation issues
A symptom of these issues is that the CCS UI becomes unresponsive. Investigation revealed that the compile/link cycle completed successfully .. but the the CCS UI was hung.
Further investigation revealed that Eclipse has long standing issues with the console being sluggish.. and it blocking when wrapping lines...

https://bugs.eclipse.org/bugs/show_bug.cgi?id=11601
https://bugs.eclipse.org/bugs/show_bug.cgi?id=519125

These issues are minimized by turning off console line wrap.

The settings for console line wrap are not shown by default .. In the preferences dialog the Show Advanced Settings Button must be pressed (The button then reads Hide Advanced Settings)

debug directory - Delete it if you have slow/hung compiles

For reasons unknown CCS will get to a point where a previously working build will either no longer work, or will compile really slowly. These issues can be often resolved by deleting the debug directory with the project folder.

This will typically happen for no obvious reason. Factors that seem to encourage this are things like shutting down CCS's host environment (in our case linux). Changing the source code underfoot also seems to encourage this. (i.e using git to change code versions.

A related issue is that if files are moved in the file system CCS will not delete compiled files in the build directory. Deleting the directory resolves this issue ..

Missing Paths - Check them if you have hung compiles

Compiles will hang if include paths are not found.. this seems to be associated with a warning that a path is not found.
Sometimes these warnings are false alarms .. sometimes not..

RSE - make sure it is disabled

As previously mentioned sometimes a build will stop when RSE tries to sync with a remote file system. Ensure it is disabled in the workspace

Troubleshooting CCS Issues

CCS Logs

CCS logs are your friends. Using a tail like tool is recommended .. one with a GUI is even better. glogg works well on Linux.
The logs are stored per CCS Workspace in the directory that CCS used to load the workspace

CCS Log
WorkspaceName/.metadata..log

Build Logs
WorkspaceName/.metadata/.plugins/org.eclipse.cdt.ui/xxx.log
There is a global build log and build logs for individual projects

CCS Progress Window

The CCS Progress Window shows the running tasks, and their progress. It will show if a task is blocking.
To show it Double Click at the bottom right hand corner of the CCS UI, to the left of the two vertical lines..



CCS Speed Optimizations

Parallel builds

All else being equal and working as expected this will have the biggest impact on build times.

In CCSProject\Properties\Build\Build Settings check enable parallel builds and choose use optimal jobs

(which should be the same as the number of cpus or virtual cpus in your PC most of the time unless you're running a VM then use:
vm_configured_thread_count - 2)

Don't bother setting it to a bigger number .. it has the effect of slowing the build down .. YMMV

Memory - you can never have enough

Out of the box CCS performs garbage collection during the build process ..

Increase Initial Heap Size for JVM Xms2048 (from 512)
Increase Max Heap Size -Xmx8192m (Modulo 1024) unless you only have 8192 system memory. -Xmx6144 is also good.

Doing this seems to make CCS much more stable

edit ccstudio.ini.. path ti/ccsv8/eclipse once per CCS

install-Xms2048m
-Xmx8192m

Software Development Methods

Writing Software

Code Composer Studio is the IDE for development as it caters to the Texas Instruments platform allowing easy debugging. It will also perform the code formatting required, and report the compiler's MISRA checking of code style, and it has it's own static code analysis reporting. All code will be generated manually with the exception of some testing framework which is able to be auto-generated in the beginning and later updated manually.

The DC Meter software folder structure will be as depicted in the figure below. Sub-modules will have their own src and test folders. This allows the modules to easily be extract and re-used in other projects keeping all related code and tests with it:

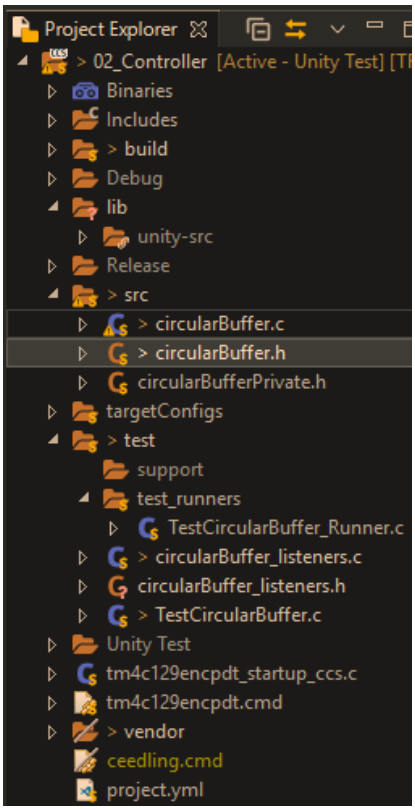


Figure - Project Folder Structure

Testing Software

Unit testing may be performed manually or automatically by the framework depending on the developer's preferences at the time. There will be a continuous integration server, which will perform a combination of nightly on target testing and off target testing.

Automated Testing

On Target Testing Vs PC/Server

Testing Metrics to Report

Code Version System

instructions for setting up Git can be found here: Git for Tritium Engineering Use. If you're using Linux then further instructions can be found in the "TRI70 /00 - Git Setup/install-prepare-commit-msg.ubuntu.sh" script.

Git will have some hooks added to allow auto testing of comments and code before permitting a commit, which is done as part of the Git preparation.

The Git repository for containing the actual DC Meter software/firmware is open to store anything. One folder will be the actual DC Meter software and everything else can be anything the project needs such as example projects, de-risking activities, documentation, etc.

Git has the command 'rebase' as a mechanism to pull from where you branched from and make all your changes happen after that rebase. For the DC Meter project, 'rebasing' is **NOT** encouraged. Git branch trees are good at showing when a change was started, how long it took, and how a merge may impact prior merges/functionality. There is also the potential for other problems when 'rebasing' especially around previous pushed commits and merging.

Prior to merging updated code back to the mainstream, a 'pull request' will be performed so a review of the update can be performed. The results of the pull request in git shall be emailed to the person responsible for performing the review and archived.

Frequently Used Git Commands

how-to: example for comparing a specific local file with the origin version of the same file. This is useful if you wanted to perform a diff prior to performing a pull. Note, difftool is used instead of diff to take advantage of 3rd party diff GUI's:

```
git difftool origin/master -- source/tri110-dcmeter-freertos-lwip2/src/cloud_task.h
```

how-to: example for comparing a specific local file of a previous version /revision with the origin, where the revision is represented as the short hash of the commit:

```
git difftool origin/master -- source/tri110-dcmeter-freertos-lwip2/src/cloud_task.h 2a03dd6
```

how-to: example list only the files that have changed for in the comparison:

```
git difftool --name-only origin/DCM-Demo -- source/tri110-dcmeter-freertos-lwip2/src/cloud_task.h 2a03dd6
```

how-to: example compare a file on one branch with the same file on another branch:

```
git difftool origin/DCM-Demo origin/TDM-366 -- source/tri110-dcmeter-freertos-lwip2/src/cloud_task.h
```

how-to: check which branches have been merged into master (the '-a' includes remote branches):

```
git branch -a --merged
```

how-to: remove a file from git that has wrongly commited and is now part of Git tracking:

```
git rm --cached <file_name>
```

how-to: merge a branch into your branch will full control over how the merge is performed:

```
git merge --no-commit --no-ff -s recursive -X patience 03728451c7099035c06c65f669b1af1656cf24e2
```

how-to: create a pull request via command line. This example requires you to have pushed your changes back to the server and the goal is to merge TDM-2104 to master.

```
git request-pull remotes/origin/master ssh://git@git.tritium.com.au/TRI110 remotes/origin/TDM-2104 >
PullRequest_TDM-2104.txt
Email the file PullRequest_TDM-2104.txt with the subject "[Pull Request] TDM-2104" to the person responsibe for
approving the pull request
```

how-to: create a pull request via TortoiseGit:

TortoiseGit has a button after the 'push' dialog, which creates a pull request.

Branching strategy

For the DC Meter project, the branching strategy used will be the feature branching strategy. In a nutshell, create a branch for the JIRA issue/task/feature you're going to work on. Do your changes. Merge back into master. For further information see: https://www.atlassian.com/git/tutorials/comparing-workflows/feature-branch-workflow

Guide to Commit Comments

The very first entry in a Git comment is the name of the branch you're working on. If you have setup Git properly, then the pre-commit hooks will do this for you. Some Git GUI applications will show you this value before you add your commit message, others will insert it after you've completed your message and you won't see it until the commit is done. Trust your GUI.

Documenting Code

Documenting code with Doxygen notation will allow auto-generation of code-documentation.

It will be up to the developer to maintain software design diagrams/documentation to keep in line with the implementation and vise versa.

Managing the release branch

Pre-requisites:

1. A JIRA ticket exists for the work
2. The JIRA ticket is assigned to a release

Merge Process (This process becomes more automated when you move to Bitbucket with its pipelines):

1. read pull request to verify
    a. only expected files were modified
    b. diff files to check modification
2. check some tests exist for the new code
3. merge code with the following command: git merge --no-commit --no-ff -s recursive -X patience <git hash of the branch to merge>
4. check differences between current and master prior to merge (maybe limit to specific files)
5. check differences between current and branch merged (maybe limit to specific files)
6. make sure it compiles in debug, release, UNITY_TEST (re-index, rebuild)
7. run in debug (no bootloader)
8. run in release with a bootloader
9. run unit tests after building for UNITY_TEST
10. checkout under Linux and build for CATCH2 and run tests (unless you get CATCH2 to run on target)
11. commit and push merge
12. ensure the related JIRA ticket is closed (consider if some comments need to be added for historical reasons)
13. update customer facing release notes (internal release notes are generated automatically)

Standards for Software Products

The VDE[R1] refers to WELMEC 7.2[R2] as a guide for development although not making it a requirement. Where possible, the software development process will comply to WELMEC 7.2[R2] as this also means we comply with MID requirements and should make certification easier.

The original requirements will not be captured in any other document than the original. Derived requirements will be captured in JIRA as tasks with a comment that links them to the parent requirement in the originating document. No formal standard is being followed for capturing requirements. In the absence of any formal guide, refer to https://confluence.atlassian.com/jirakb/using-jira-for-requirements-management-193300521.html combining JIRA and Confluence to manage requirements.

Coding style

Tritium have a formal document for this titled POL.003 -- Tritium C Coding Styles & Guidelines[R4]. Missing elements from Tritum's coding standard can be covered by:

1. https://wiki.sei.cmu.edu/confluence/display/c/SEI+CERT+C+Coding+Standard
2. https://wiki.sei.cmu.edu/confluence/pages/viewpage.action?pageId=88046682
3. https://isocpp.org/wiki/faq/coding-standards

The following deviations from POL.003 will be made:

Table - Tritium Coding Standard Deviations

POL.003 Section	Deviation	Rationale
2.2 - File description comment block	Style shall be changed to match Doxygen (http://www.doxygen.nl/manual/index.html) preferring to use the Qt style notation	Allow automatic extraction.
2.3 - Function description comment	Style shall be changed to match Doxygen (http://www.doxygen.nl/manual/index.html) preferring to use the Qt style notation	Allow automatic extraction.
2.5 – Code tags	Style shall be changed to match Doxygen Special Commands. Optionally use Markdown support to highlight existing tags	Supports automatic extraction.

Templates for Code Composer Studio can be found at `S:\Texas Instruments\CodeComposer Studio - Offline installer\Templates`.

Import existing Tritium Coding style by:

1. 'right-click' the project and select 'properties'
2. expand to: C/C++ General Formatter
3. click on "Configure Workplace Settings"
4. Click on "Import..."
5. Browse to the saved file "S:\Texas Instruments\CodeComposer Studio - Offline installer\Templates\Tritium_C.xml"
6. Click 'Apply'

To enable Doxygen support in CCS:

1. 'Right-Click' the project and select 'properties'
2. expand to: C/C++ General
3. 'Check' "Enable project specific settings"
4. Choose Documentation tool "Doxygen"
5. Click on 'OK'

To import code templates:

1. Open WindowPreferences
2. Expand to: C/C++ Code Style Code Templates
3. Click on "Import..."
4. Browse to: "S:\Texas Instruments\CodeComposer Studio - Offline installer\Templates\TritiumCodeTemplates.xml"
5. Click 'Open'
6. Click 'Apply'

The C and C++ language dialect used in writing software for the DC Meter are 'C11' and 'C++11'.

Incorporated Reusable Software Products

To reduce development time and help improve quality, some already existing and tested libraries will be used:

Table - Third Party Libraries Used in the DC Meter

Library Name	Version	License	Supplier	Description
FreeRTOS	8.2.3	MIT	https://www.freertos.org/	Real-time operating system kernel.
mbedTLS	2.13.0	Apache 2.0	https://tls.mbed.org	Elliptic curve implementations, random bit entropy generator, TLS support, signing /verification support.

lwIP	2.1.0	BSD	https://savannah.nongnu.org/projects/lwip/	Lightweight TCP/IP stack.
decNumber	0.1.1	Apache 2.0	https://github.com/amzn/ion-c	Decimal32 (base 10) arithmetic library to provide more accurate than float32/64 math operations.
Tivaware	2.1.4.178	Open	http://www.ti.com/tool/SW-TM4C	TivaWare library supporting the chosen processor. These provide access to the processor peripherals.
sts-2.1.2			https://csrc.nist.gov	Library to evaluate/test random number generation. This is used primarily as part of testing during development although some code may be used at run-time.
json-develop	3.7.0	MIT	https://github.com/nlohmann/json	Library to implement JSON, binary JSON, CBOR for configuration and messages between DC Meter and TTI
Unity/CMock/Ceedling		MIT	http://www.throwtheswitch.org	Testing framework for C
Catch2	2.6.1	Boost v1	https://github.com/catchorg/Catch2	Testing framework for C++
mqtt-c	3.1.1	MIT	http://docs.oasis-open.org/mqtt/mqtt/v3.1.1/os/mqtt-v3.1.1-os.html	Library for performing MQTT communications

Developed Reusable Software Products

Table - List of easily reusable modules for other products/projects

Software Module Name	Description
Bootloader	
crypto interface (SHA256, AES128/256, RBG, TLS 1.2, X.509, ECDSA)	
Ethernet over CAN	
BlockDevice, BlockFile, TritiumFiles	
Circular Buffer	
Software Downloader (aka updater)	
Remote Debug Printing	
Memory Allocators for STL and std::string	

Handling of Critical Requirements

The VDE[R1] addresses the requirement categories below either directly or by referring to other documents.

Safety Assurance

Safety is addressed through hardware requirements. The DC Meter is connected to high voltage, high current sources, but then fully enclosed to prevent physical access. The DC Meter software doesn't control anything safety related other than prevent a charge session starting if the log is full. There aren't any software requirements regarding safety assurance.

Security Assurance

Security assurance is addressed through either secure communications or physical restriction through tamper evident means.

Software Quality Assurance

The DC Meter software needs to be certified by the external body PTB. The following documents were used to determine what the certification requirements are.

Table - List of sources for the Product Requirements

Source	Actions	Enforceability (Guide/Strict)	Notes
"Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.005 PTB APPLICATION\QA Module D application and Research\PreliminaryList.pdf" – Preparation for a procedure to approve a quality system	ISO 9001:2000	Strict	For Harmonised Standards the example of OIML R49 is provided. However the VDE is to be our standard.

	Harmonised Standards, Normative Documents	Strict	Similarly for Directive 2004/22/EG (MID) the VDE is to be our standard as per the statement in: VDE, Appendix B.2 General: *"(mutatis mutandis, as no DC meters are covered)"*.
"Z:\Tritium\TRI110 - DC Meter\00 - Documents\TRI110.005 PTB APPLICATION\QA Module D application and Research\Criteria_adn_procedures_QS-Approval_March2018.pdf"	Self Conformity Assessment Document	Strict	This is the manufacturer's conformity assessment. Each requirement needs to be addressed with a description of how we conform to meet the requirement. This is partnered up with a Conformity Declaration.
	Conformity Declaration	Strict	
REA 6A	Product Requirements	Strict	
VDE refers to WELMAC 7.2 "Z:\Tritium\TRI110 - DC Meter\00 - Documents\Research\WELMEC GUIDES\SOFTWARE GUIDES\Guide_7.2_2015__Software.pdf"	Software Design Document	Strict	Must meet WELMEC 7.2 requirements: P1, P3, P4, P5, P7, L1, L2, L4 • notes on accuracy of measuring algorithms • description of the user interface • software identifiers and how they're constructed and different per version • system hardware overview • Data flow diagrams (comms and ui) • Protection against unintentional changes • Configuration parameters (set and secured) • Modes/states (relating to configuration changes) • Description of the stored data (usage, events, etc)
	Operating Manual	Strict	
	Source Code	Strict	

Addressing WELMEC 7.2

To assist in understanding what requirements apply to the DC Meter in WELMEC 7.2[R2], the DC Meter needs to be assigned a class. As the DC Meter is *" a built-for-purpose instrument with embedded software"* the WELMEC 7.2[R2] requirement type 'P' is applicable. The DC Meter also has functionality and hardware to support *"long term storage of measurement data"*, and *"transmission of measurement data"* and *"download of legally relevant software"* so the IT configuration (modular extensions) applicable include 'L', 'T', 'S', and 'D'. As this is an electricity meter the instrument specific extension of 'I.3' is applicable as described in Chapter 10.3 of WELMEC 7.2.

In accordance with P2/U2 we address detection of specific errors using recommendations from *WELMEC, Chapter 10 Extension I: Instrument Specific Software Requirements* as detailed below.

Important notes to consider when addressing WELMEC 7.2 concerns:

1. Some fault detections are possibly in hardware (e.g. MCU FaultISR, external flash Error Correction Codes)
2. Faults not detected by hardware are detected by software using either CRC or SHA256 hash calculations, or cryptographic signing.
3. Software detection is done regularly
4. The DC Meter is a dedicated instrument with an embedded system subject to legal control
5. The DC Meter depends on a network connection to the internet via a network switch providing firewall protection
6. The sensor is integerated
7. The user interface is always under legal control
8. There is long term, fixed (non-removable), local storage for audit logs
9. Uses the FreeRTOS operating system
10. The only other application residing on the system is the bootloader, which itself is part of the legally relevant software but separate to the metering software. Only one can run at a time, the only way to start the bootloader is to reboot, then it starts the metering application.
11. All software on the DC Meter is subject to legal control.

Using the risk table specified in the WELMEC 7.2[R2] guide in section I.3.6 the DC Meter has been identified in the risk class of **TBD**.

Table - WELMEC 7.2 Risk Class Definition

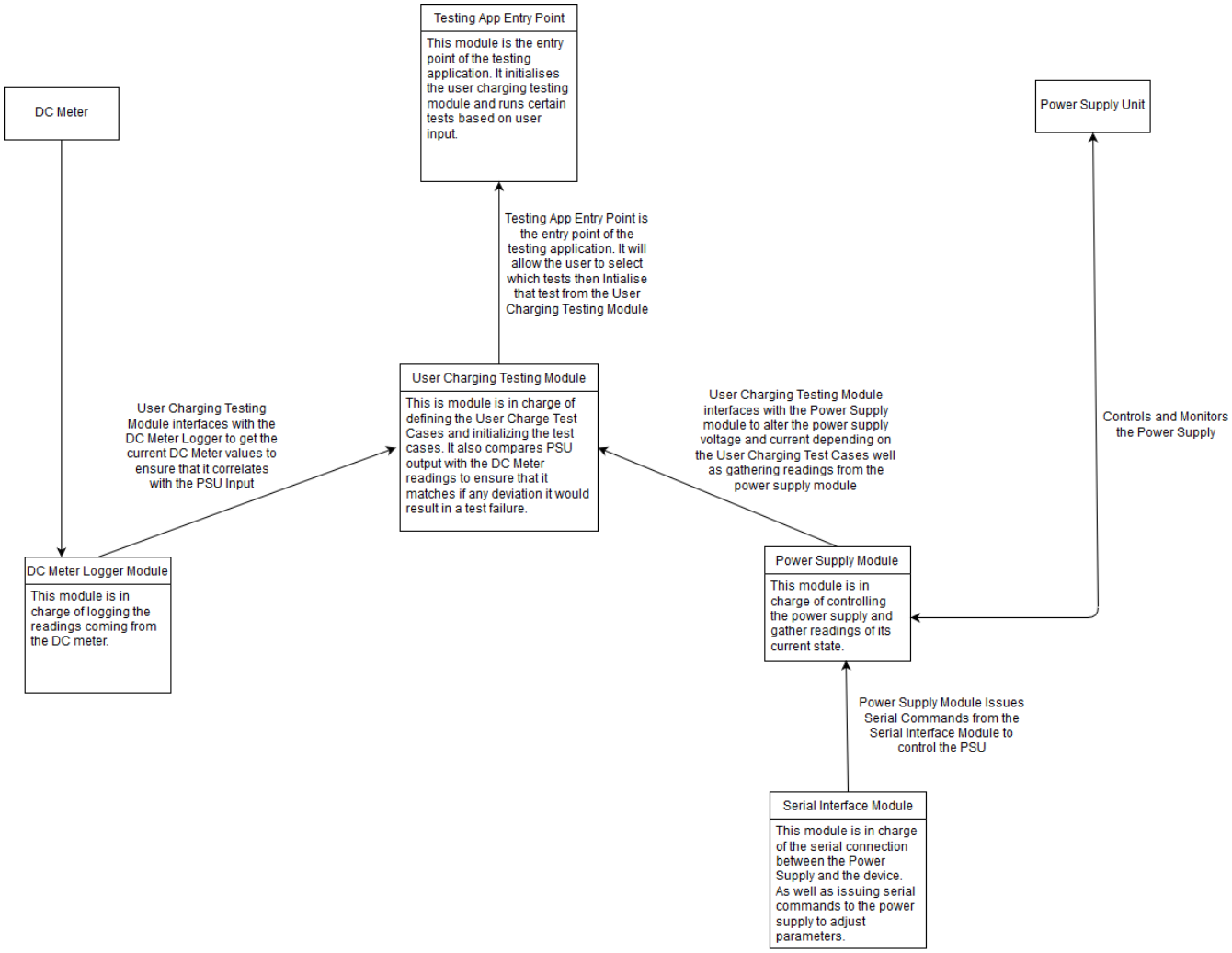
Risk Class	Software Protection	Software Examination	Software Conformity
A	Low	Low	Low
B	Middle	Middle	Low
C	Middle	Middle	Middle
D	High	Middle	Middle
E	High	High	Middle
F	High	High	High

Table - WELMEC 7.2 Risk Assessment

WELMEC 7.2 Requirement	Risk Assessment				Notes
	Software Protection	Software Examination	Software Conformity	Risk Class	
P1: Documentation					
P2: Software Identification					
P3: Influence via user interface					
P4: Influence via communication interface					
P5: Protection against accidental or unintentional changes					
P6: Protection against intentional changes					
P7: Parameter protection					
L1: Completeness of measurement data stored					
L2: Protection against accidental or unintentional changes					
L3: Integrity of data					
L4: Authenticity of measurement data stored					
L5: Confidentiality of keys					
L6: Retrieval, verification, and indication of stored data					
L7: Automatic storing					
L8: Storage capacity and continuity					
T1: Completeness of transmitted data					
T2: Protection against accidental or unintentional changes					
T3: Integrity of data					
T4: Authenticity of transmitted data					
T5: Confidentiality of keys					
T6: Handling of corrupted data					
T7: Transmission Delay					
T8: Availability of transmission services					
S1: Realisation of software separation					
S2: Mixed indication					
S3: Protective software interface					
D1: Download mechanism					
D2: Authentication of transmitted software					
D3: Integrity of downloaded software					
D4: Traceability of legally relevant software download					

Automated Test Server

Testing Application System Architecture Diagram



Controlling Diagnostic Messages

Overview

The C/C++ compiler provides diagnostic options to control compiler- and linker-generated diagnostic messages. The diagnostic options must be specified before the --run_linker option.

--diag_error=num	Categorises the diagnostic identified by num as an error. To determine the numeric identifier of a diagnostic message, use the --display_error_number option first in a separate compile. Then use --diag_error=num to recategorize the diagnostic as an error. You can only alter the severity of discretionary diagnostic messages.
--diag_remark=num	Categorises the diagnostic identified by num as a remark. To determine the numeric identifier of a diagnostic message, use the --display_error_number option first in a separate compile. Then use --diag_remark=num to recategorize the diagnostic as a remark. You can only alter the severity of discretionary diagnostic messages.
--diag_suppress=num	Suppresses the diagnostic identified by num. To determine the numeric identifier of a diagnostic message, use the --display_error_number option first in a separate compile. Then use --diag_suppress=num to suppress the diagnostic. You can only suppress discretionary diagnostic messages.
--diag_warning=num	Categorises the diagnostic identified by num as a warning. To determine the numeric identifier of a diagnostic message, use the --display_error_number option first in a separate compile. Then use --diag_warning=num to recategorize the diagnostic as a warning. You can only alter the severity of discretionary diagnostic messages.

Suppressed Warnings (3rd Party Libraries)

The DC Meter uses 3rd party libraries to deliver certain functionality. 3rd party libraries are evaluated to be fit-for-purpose and are not maintained by Tritium. Warnings generated from these 3rd party libraries are seen as beign and are suppressed.

MQTT Library

#112-D Statement is unreachable.

#190-D Enumerated type mixed with another type.

Doxygen

Introduction

Doxygen is being used as the tool to generate documentation for the Source Code for the DC Meter. Doxygen allows C & C++ code (among others) to be parsed and to generate

HTML (among others) files that document the source code.

Source Code can be annotated with Doxygen Commands to help in this process. See http://www.doxygen.nl/manual/commands.html for details.

Doxygen distribution

Doxygen is distributed as part of Ubuntu. The distributed version atm is 1.8.13

Doxygen may also distributed as part of tool chains that use it. This saves having to integrate it into the host OS.

Doxygen may be downloaded, compiled and installed to use the current version.

Building, Installing Doxygen on Ubuntu

The current version of Doxygen is 1.8.18

See http://www.doxygen.nl/download.html for instructions for downloading, building and installing the source from git

When installing on the UbuntuVM cmake, flex and bison may also needed to be installed.

Doxygen and Code Composer (CCS) / Eclipse

Out of the box both Eclipse and Doxygen allow the selection of which version of Doxygen to use amongst those that are installed on the host.

CCS/Eclipse integrate highlighting of Doxygen Commands in several different formats.

See http://www.doxygen.nl/manual/docblocks.html for details.

CCS/Eclipse will auto generate a formatting block for a function when the following text is typed (in bold italics) above a function declaration

//!<enter>

```
//! @param aaa
//! @param bbb
```

//! @return

/*!<enter>

/*!

*

 * @param aaa

 * @param bbb

 * @return

 */

/// <enter>

/// @param tariffDescriptor

/// @param theTime

/// @param elapsedTimeMinSec

/// @param meteredEnergyThisSession_kWh

/// @return

Note that the @param and @ return format are fixed. There is no easy means to customise them.

See https://stackoverflow.com/questions/4403192/change-doxygen-comment-style-in-eclipse for details

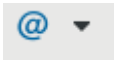
Doxygen and Eclox

Neither Eclipse or CCS are capable of running Doxygen out of the box to generate documentation.

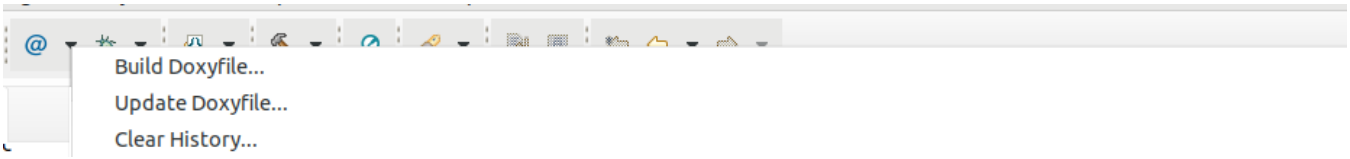
A third party add in, Eclox is used.

See https://anb0s.github.io/eclox/ for details

Eclox bundles a binary version of Doxygen (1.8.13??)

After installation the only hint that it has installed is the appearance of a Blue @ in the toolbar. 

Pressing the drop down arrow shows



Eclox allows the creation and modification of .doxyfile files.

A .doxyfile can be initially created by right mousing on a folder within CCS/Eclipse and choosing New\Other\Filter on dox then creating the file.

Once the file has been created double click on it to open the .doxyfile editor (courtesy of Eclox)

Project setting can be set on both the Basic and Advanced page.

ToDo .. come up with preferred settings ...

Doxygen and the DC Meter Code Base

The DC Meter is a mix of C and C++ Code. A recurring pattern is for a C++ class to have associated C code in the same header file.
The DC Meter uses .h for header files for both C + C++ code.

Doxygen is not a based on a compiler, it parses the code independently.

When a C++ compiler is compiling a file the macro__cplusplus is defined. When a C compiler is compiling a file the __cplusplus macro is not defined.

This is used in the header file to conditionally compile the C and C++ code by testing if __cplusplus is defined or not.

Doxygen does not do this, so consequently when a file that uses this technique is parsed by Doxygen the C code is parsed, and not the C++ Code.

There are a couple of workarounds for this.

One is to use Doxygen filters, which is a mechanism to pipe a file through a filter to preprocess it.

An alternative is to redefine a macro within Doxygen. This is the easier (for now) solution.

The Macro DOXYGEN is defined within Doxygen, and the following text in added to the top of header files that have a mix of C & C++ Code.

```
// When Doxygen is parsing ensure __cplusplus is defined

#ifdef DOXYGEN

#define __cplusplus

#endif //DOXYGEN
```

To define the DOXYGEN macro open the .doxyfile, go to the Advanced Dialog, Custom Tab. Search for Predefined Setting.

Predefined DOXYGEN=1

Doxygen vs Tritium C Coding Styles & Guidelines

In order to get the best out of Doxygen some of the Style Guidelines require a reinterpretation .. (Regard this as a process .. of which this is the first step ..)

2.1 Variable Names in comments : Doxygen explicitly uses Variable Names as part of a comment block. Once inserted there is no mechanism to update the comment if the variable name changes ..

```
//!
//! \param[in]          param1
//! \param[out]         param2
//! \param[in/out]      param3
//! \result             describe result
//! \brief              one line description of function
//! \details            details on separate line
//! \details            more details on separate line
//!
```

2.1 File Description Comment Block : If a \brief description, and then \details, are added under the copyright in an implementation file they are treated as documentation for the first thing that is declared in the file.

To associate with a specific class tag the text with \class

```
//! \class ClassName
//! Some Text
//! More Text
//! More Text
```

Extensions to the Tritium C Coding Standard

- Introduction
    - References
        - Web Links:
        - Books:
- Rules

Introduction

Here are some items not covered by the Tritum C coding standard that I think we should address for consistency sake and clarity. I encourage discussion of Tritium's coding standard with the goals of understanding it, improving it and us as developers, use it as a platform for:

1. better code-reuse,
2. more efficient with time on design, implementation, and testing,
3. better reliability

References

Web Links:

http://wiki.c2.com/?CodingStandardList
http://www.gnu.org/prep/standards/
https://isocpp.org/wiki/faq/coding-standards
https://users.ece.cmu.edu/~eno/coding/CppCodingStandard.html
https://wiki.sei.cmu.edu/confluence/display/c/SEI+CERT+C+Coding+Standard
https://wiki.sei.cmu.edu/confluence/pages/viewpage.action?pageId=88046682
https://www.approxion.com/capital-offenses-how-to-handle-abbreviations-in-camelcase/
https://stackoverflow.com/questions/15526107/acronyms-in-camelcase
http://cppunit.sourceforge.net/cppunit2/doc/coding_guidelines.html
http://wiki.c2.com/?CamelCase
https://isocpp.org/wiki/faq/mixing-c-and-cpp

Books:

"C++ Coding Standards: 101 Rules, Guidelines, and Best Practices". Herb Sutter and Andrei Alexandrescu, 8th August 2010.

Rules

Names containing acronyms when using camel case naming scheme often leads to an abomination in inconsistent names. For some it seems clear but adding some clarification to avoid confusion is a relatively simple task. Below I've identified what I think are the appropriate clarifications but I'm open to other suggestions. Unfortunately for me I have to change what was drummed into me at BAE Systems.

Basic Camel Case Capitalization Rules according to online references. This is provided as a starting point and does not take into account Tritium's coding standard.

1. The first letter is capitalized.
2. One or more letters in that word are also capitalised.
3. The word does not end on a capitalized letter: CamelCasE
4. No two capitalised letters shall follow directly each other: CamelCAse
5. No number in that word at any place: CamelCase1more
6. No dot(.), under_score or dash (-) within the word, only letters: Camel_Case
7. No ‘foreign’ letters in it like äöüß or accentuated like áéí. CämélCáße

Deviations for Tritium's coding standard:
point #1 needs to be changed to: "The first letter is capitalised only for Enumerations and Structures. Functions, variables (of any scope), function arguments, and file names, shall all begin with a lowercase letter."

Extensions to these rules to cater for ambiguities in dealing with abbreviations and acronyms in an attempt to better fit with Tritium's coding standard
8. Rules #4 and #6 shall prevail therefore acronyms and abbreviations shall be treated the same as words with the exception of units (see rule #9) and identifying the location of the function/global variable according to rule #9.1
8.1. If the acronym begins the name then it starts lower case and all acronym letters are lower case
8.2. If the acronym is mid name, it starts with an uppercase letter and remaining letters a lower case
8.3. If the acronym consists of only two letters, then capatilise both and try to place at the end unless rule #9 applies. (e.g. IO looks getter than Io but uV is proper)
9. Rule #6 can be broken under the following conditions:
9.1. to avoid name clashes in 'C', public functions and global variables (non-static) must be prefixed with a series of letters representing the file name they are in followed by an under_score. e.g. myFile_myGlobalVariable
9.2. to easily support units in names, the unit is postfixed to the name separated by an under_score with capitalisation conforming to SI guidelines (e.g. myfile_sampledData_mV).
9.3. the first letter following an under_score must be lower case unless it is an SI unit which is uppercase (e.g. V for Volts)
10. the first letter of the first word of the name is lowercase

Usage of the 'extern' keyword.
1. Whilst extern is automatically assumed in header files by the compiler so doesn't need to be explicitly stated except for clarification when the implementation is not in the 'c' file for that header.
2. In 'c' files, extern should be used to define functions if they're implemented outside that file and for modularity/portability/adaptability sake it is better to avoid including the header file. e.g. only the function API's are needed. This leave flexibility for the function to exist anywhere.

"#include" style.
two include styles exist:
#include <file>
#include "file"
They are different in that it tells the compiler where to start looking for the file. The first is for libraries and the second is for code you added. You can still reach a library header using double quotes as the library path is checked second. The former should not be used for code you added.

If your module allows for configuration, and that configuration can be done using #defines in a header file.
e.g. #include CONFIG_HEADER_FILE
elsewhere: #define CONFIG_HEADER_FILE "theConfigFile.h"
or in project settings -DCONFIG_HEADER_FILE="\"theConfigFile.h\""

Is it more important to enforce a strict coding layout across a company, or to be consistent within the file? Would it matter if there was a process to automatically format code on a commit hook using a tool like clang?

For a coding standard (not style guide) I recommend the Secure Coding Standard (https://wiki.sei.cmu.edu/confluence/display/c/SEI+CERT+C+Coding+Standard) over MISRA but from a company strategic direction we need to fill the gaps and address the concerns of the wider community regarding security as well as reliability.
For C++ I recommend the minimum of https://isocpp.org/wiki/faq/coding-standards and any of the 3 links to valid standards and recommend https://wiki.sei.cmu.edu/confluence/pages/viewpage.action?pageId=88046682.

Memory management needs to be consistent and it isn't always using the default malloc/calloc/free. So how do you neatly specify different memory management functions?

To avoid deeply nested if/else conditions and multiple return statements, wrap code in a do/while(0) and use break to exit.

Where to use the #include when mixing C and C++ code: as per https://isocpp.org/wiki/faq/mixing-c-and-cpp system headers are included outside the extern "C" and 'our' headers are inside. This also restricts the scope to just what is in the header that was created by 'us' instead of what is in the included headers. There may be a subtle side-effect otherwise.

Initialisation of structs at declaration should use use the ".member = value" notation to improve immunity during maintenance against change in the structure layout. new fields will be auto initialised to 0. Helps with self-documenting code. Prevents dependency on ordering of field initialisation. e.g)

```
struct myFile_sampleStruct
{
        uint32_t fieldA;
        uint8_t fieldB;
};

const struct myFile_sampleStruct sampleStruct =
{
        .fieldA = 1234;
        .fieldB = 56;
};
```

when an 'if' statement contains an immutable value as part of the condition, ensure the immutable is first in the condition. This helps prevent accidents where an '=' is missed from an equality check making it an assignemnt because the compiler will tell you it is an error, as opposed to the assignment taking place and if the result is == 0 or not.
e.g. "if(IMMUTABLE_VALUE == variableValue)" instead of "if(variableValue == IMMUTABLE_VALUE)"

Style Guide:

The following rules of thumb apply to any layout chosen. If in doubt ask these questions:
1) Code should be written once and read multiple times, so make it easy to read.
2) Be wary of fitting too much on one line. Typically people read in columns.
3) Avoid syntax that can lead to common problems that are difficult to find but easy to avoid
4) Code is typically easier to read when formatted in columns.

MQTT Broker Notes

Unsecure (port 1883)

.\mosquitto_sub.exe -t 'test/topic' -v

.\mosquitto_pub.exe -t 'test/topic' -m 'Host hello world!'

.\mosquitto.exe

Secure (port 8883)

.\mosquitto_sub.exe -t 'test/topic' -v --cafile "C:\Program Files\mosquitto\certs\ttiServerCA.crt" -p 8883 --insecure

.\mosquitto_pub.exe --cafile "C:\Program Files\mosquitto\certs\ttiServerCA.crt" -p 8883 -t 'test/topic' -m 'Host > hello world!' --insecure

.\mosquitto.exe -c mosquitto.conf

Getting the SAFE Certificate

.\mosquitto_sub.exe -t 'DCMS/CSMS/SAFE' -v --cafile .\dcmRoot-mqtt.crt -p 8883 --insecure

At end of charge, you get a data dump like this:


```
DCMS/CSMS/SAFE {"SessionID":"","EDSI":"0000000104","SAFE":"
PD94bWwgdmVyc2lvbj0iMS4wIiBlbmNvZGluZz0iVVRGLTgiPz4KPHZhbHVlcz4KCTx2YWx1ZT4KCQk8c2lnbmVkRGF0YSBmb3JtYXQ9Ik9DTUYi
IGVuY29kaW5nPSJwbGFpbiI+T0NNRnx7IkZWIjoiMS4wIiwiR0kiOiJUcml0aXVtIFB0eSBMdGQiLCJHUyI6IjEyMyIsIlBHIjoiVDAwMDAwMDAx
MDQiLCJNViI6IlRyaXRpdW0gUHR5IEx0ZCIsIk1TIjoiMTIzIiwiTUYiOiIxLjAwOSIsIklTIjoiZmFsc2UiLCJJTCI6IkhFQVJTQVkiLCJJRiI6
WyJSRklEX05PTkUiLCAiT0NQUF9OT05FIl0sIklUIjoiTE9DQUwiLCJJRCI6IiIsIlZTIjoiUlhaWU1YTVJQRVNBIiwiUkQiOlt7IlRNIjoiMjAy
MS0wOS0xNVQwNzo0ODowMCwwMDArMDIwMCBTIiwiVFgiOiJCIiwiUlYiOiIwLjAwMCIsIlJVIjoia1doIiwiU1QiOiJHIiwiUkkiOiIxLWI6MS44
LmUiLCJVVCI6IiAifSx7IlRNIjoiMjAyMS0wOS0xNVQwNzo0ODo0MywwMDArMDIwMCBTIiwiVFgiOiJDIiwiUlYiOiIzLjk1OSIsIlJVIjoia1do
IiwiU1QiOiJHIiwiUkkiOiIxLWI6MS44LmUiLCJVVCI6IiAifSx7IlRNIjoiMjAyMS0wOS0xNVQwNzo0ODo0MywwMDArMDIwMCBTIiwiVFgiOiJF
IiwiUlYiOiIzLjk1OSIsIlJVIjoia1doIiwiU1QiOiJHIiwiUkkiOiIxLWI6MS44LmUiLCJVVCI6IiAifV19fHsiU0QiOiIzMDQ0MDIyMDZGMTdG
RTQ0NzM1MDZEMUE5Qzk4MDgyOURGREZGNjNGNDdDRTM0RTQzOTFCMjM2NkYxQThFQkQ5NjRBMzgzRDUwMjIwMjUwNUYzQzdFNEFGNkQ0MThFRDE0
NkE4QURFQUQ1NTFCOTdFOUIwMTE1ODlERTY2RDNGRDkzM0NCRTc0MDYwMSJ9PC9zaWduZWREYXRhPgoJCTxwdWJsaWNLZXkgZW5jb2Rpbmc9InBs
YWluIj4zMDU5MzAxMzA2MDcyQTg2NDhDRTNEMDIwMTA2MDgyQTg2NDhDRTNEMDMwMTA3MDM0MjAwMDQ0N0FGRkFCNjU0QzMyQjhBODAyNjg5NDU2
MDUwRTRGRjM2RDVDRTIzOTYwRjdFMjBBMDgyNUQ3REI3RjY1QjFFQTA5MDg2QURCOTEwNDdCODkxNDIwQjJEMzAxRTc0NDFDMTQxQ0ZBRUVDRTJB
M0UwOUREQkZGNDlGODgzQUMzNzwvcHVibGljS2V5PgoJPC92YWx1ZT4KPC92YWx1ZXM+Cg==="}
```


Copy the "SAFE" data string to this page: https://codebeautify.org/base64-to-json-converter

Grab the XML output - this can be pasted in to Transparency Software.

MQTT - Mosquitto

Introduction

MQTT is a lightweight protocol used to publish and messages.

Clients are able to publish data to the broker on specified 'topic/s' or channel/s.

Multiple clients can connect to a single *broker,* acting as a server which then distribute the data to clients that have subscribed to specified 'topic/s'or channel/s.

The client source library was retrieved from Eclipse Paho MQTT, and installed with 'pip install paho-mqtt'

Broker was set up using Mosquitto, no security protocol was implemented, however it can be done using the mosquitto.conf file show below.

Publish

Client publish messages on specificied 'topic', no topic configuration is required. Similar to command prompt , topics have hierachy using / as separator. E. g topic= Measurement/inputVoltage/Watts

Prior to publishing, the payload must be encoded. the function 'process_msg' in client.py can be call. It detects the type of data to be publish and encode to JSON.

Publish method require a topic, and several optional parameters:

1. qos (quality of service). Default, qos = 0, the broker does not acknowledge the message have been received and the payload is not stored and re-transmitted by the client

   qos = 1. This requires an acknowledgement from broker and client stores the message until broker send a receipt (PUBACK)

   qos = 2. Message is only received once by the broker.

For the purpose of testings qos = 0 as data is sent at short interval and few messages are allowed to get lost occasionally.

2. retain. On publishing, retain = False (default). If client publish on an unsubscribed topic, the message is lost in the broker.

   if retain = True, the last payload sent by the client will be stored in the broker and will be distribute upon subscription.

e.g client.publish(topic='Test', payload='a string, dict, list, etc' , qos= 0, retain=False)

Subscribe

A client can subscribe to multiple topics to the broker. The required parameters is topic and qos

qos = 0. Default, the subscriber may not receive the payload, however it will not be received twice

qos = 1. The payload will be received at least once by the subscriber, but the same payload may be received again

qos = 2. The payload will be received exactly once. The slowest transfer as multiple 'handshakes' are required.

A publisher with qos=2 is not guaranteed that the payload will be received by Subcriber using qos = 0.

For the purpose of testings, qos =0 is used as large constant data transmission will be received constantly by the TTI and faster speed is preferred.

e.g client.subscribe(topic='Test' , qos = 0)

Callbacks:

ON_MESSAGE

When a message is received on a topic that the client subscribed to, the class method 'on_message' is automatically called. Thus on_message is used to process received payload.

```
def on_message(self,client,userdata,message):
    msg_in= message.payload.decode('utf-8')
    msg_in = json.loads(msg_in)
    if isinstance(msg_in,dict):
        self.payload_dict = msg_in
    if isinstance(msg_in,str):
        self.payload_str = msg_in
    if isinstance(msg_in,int):
        self.payload_num = int(msg_in)
    if isinstance(msg_in,list):
        self.payload_list = msg_in
```

The payload can be retrieved by calling the class attribute. E.g if the expected payload is of dict type, it can be called with *client.payload_dict.*

Since the expected payload sent by the Charger is expected to be of <dict>, the method on_message will provide the basic functionality of deserialising JSON payload and save the payload to the class attribute payload_dict.

ON_CONNECT

When the client connect to the broker, the broker send back a CONNACK message, which in response will call the method on_connect automatically. One of the parameter is rc. Which can be called for connection acknowledgement and debugging purposes.

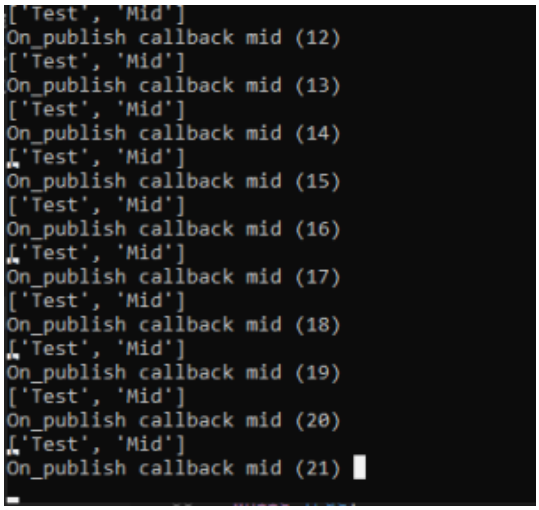
```
The value of rc indicates success or not:
    0: Connection successful
    1: Connection refused - incorrect protocol ver
    2: Connection refused - invalid client identif
    3: Connection refused - server unavailable
    4: Connection refused - bad username or passwo
    5: Connection refused - not authorised
    6-255: Currently unused.
```

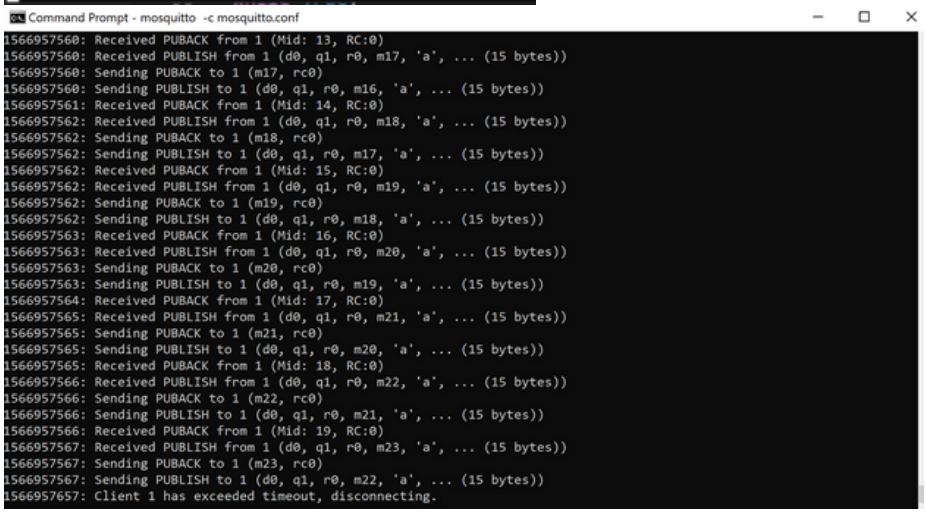
In the class method on_connect, the pass statement was used, however for debugging purposes, the rc value can be printed on the screen once the codes are uncommented.

ON_PUBLISH

Similar to on_connect, once the Client have published the payload to the broker, the on_publish callback will automatically be called. In the class method, the pass statement was used. However, in addition to rc, on_publish callback can also return 'mid' parameter

As shown above, rc == 0 signifies the message have been published successfully to the broker or else, an error have occurred. Mid value is an integer that corresponds to the published number as assigned by the client, it is essentially the message id.





Client_message_id Broker_log

By using the on_publish function, we can see that the payload transmission between broker and client is lagging. As the mid does not match between the two. The Quality of service (qos) for both client and broker

for the above logs was set at 1, which means that the payload must be received at least once by the subscriber at the cost of transmission speed. If the qos =0, the transmission speed will be faster, however the mid will always equal 0, as at qos =0, the broker and client will not set a unique payload_id.

During testing, the on_publish method simply have a pass statement, and qos is set to 0. However during debugging, qos should be set to 1 and the on_publish method should print the mid value.

ON_SUBSCRIBE

Similar to the above callbacks, on_subscribe is automatically called when the broker acknowledges. Pass statement was written here, however, the print function can be used to print the rc value.

Loop

Loop function must be called to initiate the client. If they are not called, incoming network data will not be processed and outgoing network data may not be sent in a timely fashion. There are three types of loop functionality

Loop(timeout=1.0, max_packets=1)

This is called regularly to process network events. While being called, it waits until the network socket is available for reading or writing, if appropriate, then handles incoming/outgoing data.

E.g:

client_1 = clients()

While True:

client_1.loop()

Loop_start() / Loop_stop(force=False)

These functions allow threaded interface to the network. Loop_start() must be called before connect(), then it wil runs a thread to call loop() automatically. This frees up the main thread for other work that may be blocking.

loop_stop() is called to stop the background thread.

E.g:

client_1 = clients()

client_1.connect('localhost')

client_1.loop_start()

While True:

client_1.publish('Topic',payload=None)

Loop_forever()

This is a blocking form of the network loop and will not return until the client calls disconnect. Additionally it will automatically handles reconnecting

loop_forever(timeout=1.0, max_packets=1, retry_first_connection=False)

It is recommended to simply uses loop() for testings unless threaded interface is required.

Usage:

The general usage is as follows:

1. Create client instance: client = *clients('client_id')*
2. Connect to a broker using one of the connect() functions: *client.connect('192.168.43.166' , port 1883, keepalive=60)*
3. Call one of the loop() functions to maintain network traffic flow with the broker, as shown above
4. Use subscribe to subscribe to a topic and receive messages published by the broker: *client.subscribe('Topic')*
5. Use publish() to publish messages to broker: *client.publish('Topic', payload='Test String')*
6. Use disconnect() to disconnect from broker: *client.disconnect()*

Callbacks will automatically be called upon receiving, publishing, connecting, subscribing to the broker.

MQTT - Broker set up

Installation

https://mosquitto.org/download/

After mosquitto package have been installed from the link above. Open command prompt and go to the mosquitto install directory and type *mosquitto*

You can use *mosquitto -h* for various command line switches

Mosquitto Configuration

The configuration file is included in the installation folder (mosquitto.conf). By default, MQTT Broker does not need the configuration file to run, as at default mosquitto.conf is completely commented out.

To start configuring the MQTT, edit the commented lines or create another blank config file.

Setting listener

bind_address set a specific IP address/hostname that can access the broker, if not set, the broker will be accessible to all network interfaces. In my config file, this was left empty.

port The recommended port for MQTT is 1883. For TLS the port is 8883 (This is currently not implemented). The default is 1883.

max_connections Set the maximum number of clients connection. Default is -1 which is unlimited

Logging

There are many type of messages log, which are displayed on the MQTT Broker console. Possible types are: debug, error, warning, notice, information, none, subscribe, unsubscribe, websocket and all. It is recommended that all of the log types are enable for debuging purposes.

e.g *log_type information*

Security

There are three ways Mosquitto broker verify the identity of MQTT client:

TLS

TLS can be enabled for the MQTT Broker, however it does not accept self signed certificate. Thus it was not enable for testing purposes between Charger - DcMeter and TTI.

Usernames and passwords

There is option to prompt the client to input username and password before a connection is allowed. However this is also not implemented.

Client ids

use_identity_as_clientid This replace the clientid that a client connected with with its username. This allows authentication to be tied to the clientid, which means that it is possible to prevent one client disconnecting

another by using the same clientid. For flexibility, this is set to true

```
#Define Port
port 1883

#Setting log displayed on console
log_type error
log_type notice
log_type information
log_type subscribe
log_timestamp true

#Set extra listener port
listener 8883

#Security
use_identity_as_clientid true
```

Mosquitto.conf

The Mosquitto broker does not need the config file to run. Thus it must be called explicitly to run the Broker with config file.

Ensure that the config file is in the same directory as mosquitto.exe and type the following: *mosquitto -c mosquitto.conf*

You should see the following screen if all is good:

For more information on MQTT Broker set up, https://mosquitto.org/man/mosquitto-conf-5.html

The GitHub for paho Client, https://github.com/eclipse/paho.mqtt.python

Notes on Measuring Software Performance

The documentation in this section is an informal recording of various information relating to measuring software performance as part of the development process. Any formally adopted processes will be captured in main 'Software Development Plan' page.

Table - References

Ref ID	Link/Reference	Description
PR1	https://www.freertos.org/FreeRTOS_Support_Forum_Archive/December_2011/freertos_Measuring_CPU_usage_4866946.html	Tips on using CPU usage metrics in FreeRTOS

Review of Enterprise Architect

- Introduction
- References
- Goals of the software
- Executive Summary
- Notes

Introduction

A review of Enterprise Architect (EA) from Sparx Systems was performed to see if it could help Tritium to become more efficient with time whilst improving quality and being able to supply documentation suitable for the certification requirements of the DC Meter. The information on this page contains information about how the EA software was evaluated and notes about observations of its usefulness.

References

ID	Name	Description
R1	Sparx Systems	Manufacturer of the Enterprise Architecture software found here: https://sparxsystems.com.au
R2	Tutorial Videos on EA	A useful link to videos that explain how to use EA: https://sparxsystems.com.au/resources/demos/index.html

Goals of the software

1. Easily synchronise design with implementation
   This involves generating code from design elements and the ability to round-trip code back into the design. In particular focusing on class diagrams, sequence diagrams, state diagrams.
2. Verification of state machines
   State machines are a common design element in software. Having the ability to verify state flow before implementation would greatly reduce time and improve quality
3. Ease of use
   If it takes weeks of configuration to get it right, and it feels like round-tripping is a fight every time, then maybe it isn't worth the effort.
4. Portability
   Not all software created at Tritium runs on windows and is compiled with MS Visual Studio. EA would need to do well compiling code using a cross-compiler, and work for the programming languages C and C++, assembler
5. Quality
   There must be a mechanism to ensure generated code conforms to a specified style guide or coding standard.
6. Integration with commonly used tools
   In particular JIRA, Eclipse (and hopefully Code Composer v8), MS Studio (free version), gcc compiler would be beneficial.

Executive Summary

Pros	Cons
	Doesn't run natively under Linux but will run under WINE

Notes

Version evaluated: v14, Professional

Ubuntu 18.x LTS VM- No Login Screen

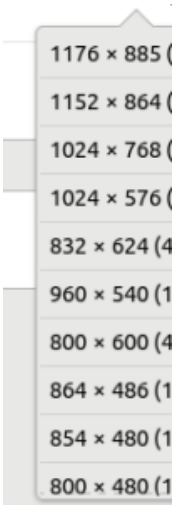
Introduction

After updating Ubuntu 18.x LTS running as a VMWare VM you are no longer able to log in to Ubuntu. Instead you are stuck on a purple screen.

Digging deeper it looks like the VMWare SVGA driver is no longer working as expected .. even though it was before the update.

It it possible to force a login by editing the grub command line and adding "nomodeset" to the linux command.
Adding the nomodeset parameter instructs the kernel to not load video drivers and use BIOS modes instead until X is loaded.
Doing this results in the login screen being shown.

Once logged in available screen sizes are limited to a maximum of 1176 x 885 pixels.



Investigation

Looking at the vmware.log shows the following entry

SVGA-GFB: Truncated maximum resolution for register modes to VRAM size: VRAM=4194304 bytes

For reasons that are not clear the SVGA driver is limiting itself to 4MB of RAM .. which is not enough to allocate screens bigger than 1176 x 885 pixels

I was able to replicate this behaviour on a clean install of Ubuntu 18 LTS .. updating resulted in exactly the same behaviour.
Whatever the root cause is is in the distro.

VMware have a flag that allows for minimum RAM allocation to be 8MB
svga.minVRAM8MB=TRUE – ensures 8Mb allocated to RAM rather than 4 .. does not go bigger than 8

Workround

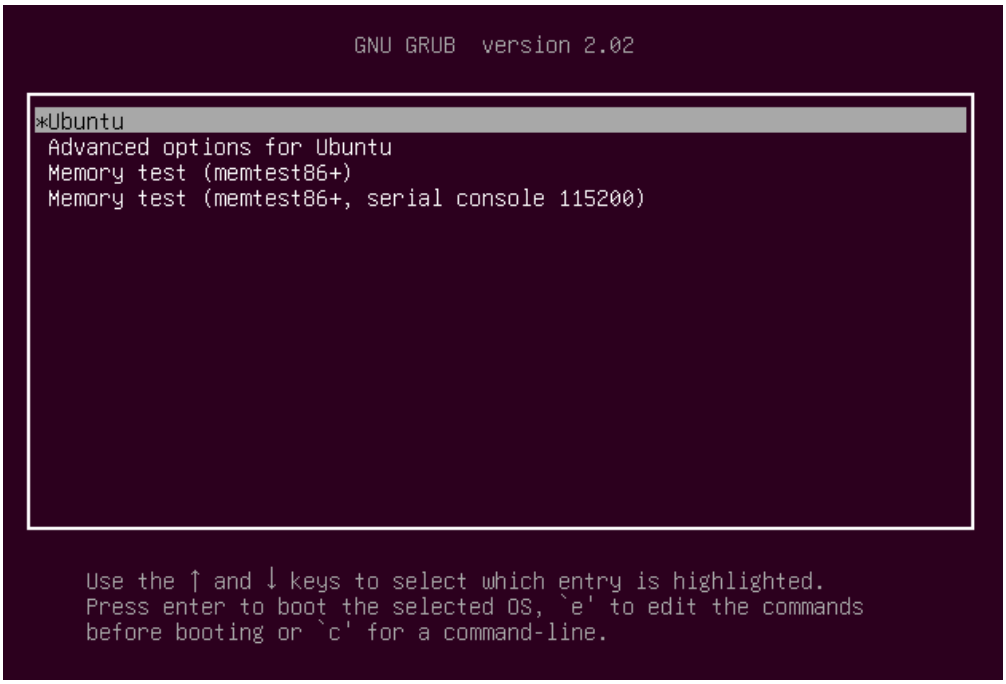
Edit the VM associated .vmx file. Do this with VMWare workstation NOT running.

Change svga.autodetect from TRUE to FALSE
svga.autodetect = "FALSE"
This will prevent the screens to be auto updated, and will force the use of config within the .vmx file

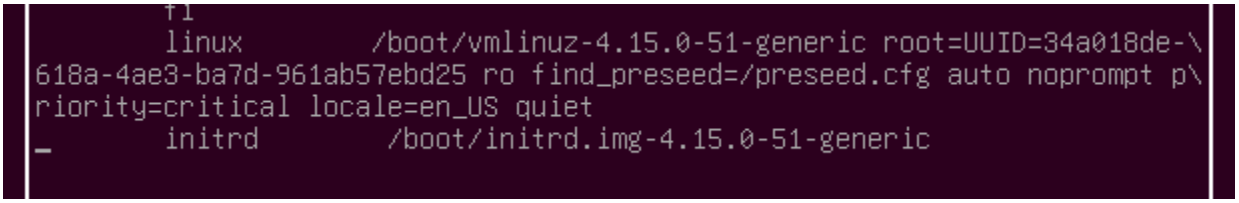
Add an entry for svga.minVRAM8MB, set to TRUE
svga.minVRAM8MB=TRUE
This will force the SVGA driver to allocate 8MB of RAM .. enough for 1920 x 1080

Add nomodeset command temporarily to grub command

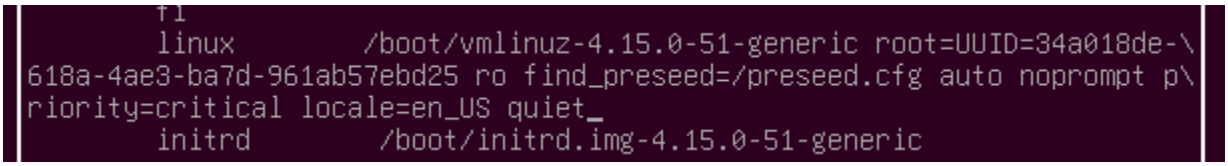
Boot VM
Click in the VM Window as it starts
Hold Down the Shift Key to get to GRUB Screen



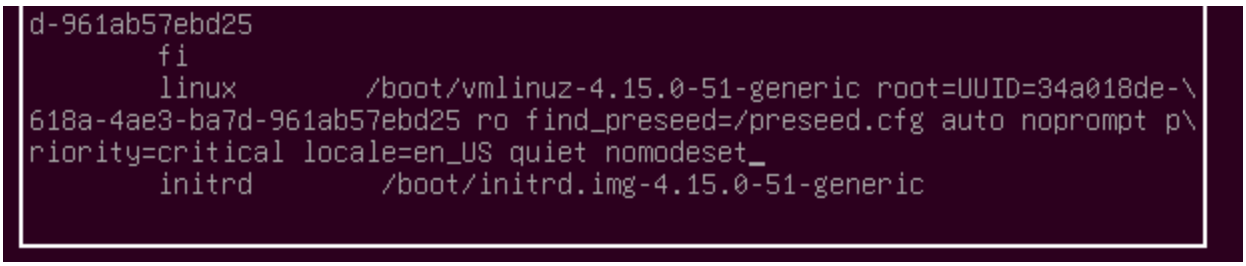
With *Ubuntu selected press e to edit grub options
Press down arrow until cursor is on line with initd on it ..



The press left arrow once .. cursor will move to end of previous line



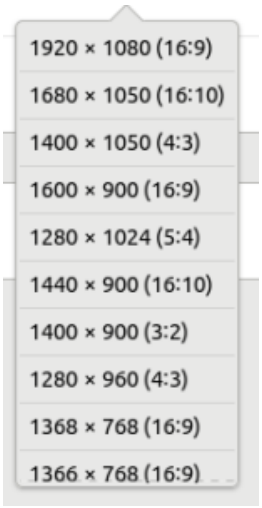
type " nomodeset"



Then press F10 key to continue boot process

This will bring up Login Screen .. Login as usual

Check screen resolutions .. expect to see resolution list like this ..



If you dont get resolutions bigger than 1176 x 885 then screen buffer is 4MB

To make nomodeset permanent

At a command prompt:

cd /etc/default
sudo nano grub

Modify
GRUB_CMDLINE_LINUX_DEFAULT="quiet"
to
GRUB_CMDLINE_LINUX_DEFAULT="quiet nomodeset"

```
GRUB_DEFAULT=0
GRUB_TIMEOUT_STYLE=hidden
GRUB_TIMEOUT=10
GRUB_DISTRIBUTOR=`lsb_release -i -s 2> /dev/null || echo Debian`
GRUB_CMDLINE_LINUX_DEFAULT="quiet"
GRUB_CMDLINE_LINUX="find_preseed=/preseed.cfg auto noprompt priority=critical l$
```

Save File
type sudo update-grub

```
dev@ubuntu:/etc/default$ sudo update-grub
Sourcing file `/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-4.15.0-51-generic
Found initrd image: /boot/initrd.img-4.15.0-51-generic
Found linux image: /boot/vmlinuz-4.15.0-50-generic
Found initrd image: /boot/initrd.img-4.15.0-50-generic
Found memtest86+ image: /boot/memtest86+.elf
Found memtest86+ image: /boot/memtest86+.bin
done
dev@ubuntu:/etc/default$
```

reboot VM .. should reboot to login screen

After making these changes the screen will maintain the resolution it has been set to .. it does not auto resize
This is expected to be a temporary workround till someone upstream notices this is broken !

X1 Prototype Hardware Issues

Design issues learnt that will be fixed on M1 prototype:

#	Item	Issue	Remedy	Status	Completed
1	Controller PCBA	eVref is connected to VDDA, Using external 2.5V reference means it won't boot.	Separate eVref and VDDA. Connect VDDA to 3v3 rail through ferrite bead.	Minor. To be done by the design engineer.	•
2	Controller PCBA	Creepage distance of 5.5mm around isolated section to AFE board does not meet pollution level 2, non potted requirements.	Decide on pollution degree of DC meter enclosure (to pot or not to pot)	Will require conformal coating. (Creepage: 3.2mm@PD1; 5.0 mm@PD2. Clearance: 3mm x2 for reinforced)	•
3	Controller PCBA	Choice of reinforced isolation or non-reinforced isolation components	Reinforced isolation will be used as reduce size and cost as appropriate components have been found	Decided.	•
4	Controller PCBA	Need to implement double isolation (or reinforced but not preferred) Consult IEC 61010. Rated to 1.2 kV requirements. Company practice should be that there is double isolation on any PCB that has both a low and high voltage connection. Each barrier needs to withstand 3kV dielectric (1 minute withstand) and 4 kV impulse.	Decide between option 1 and 2: 1 - Single isolation barrier on comms + single on AFE 2 - Double between MCU and AFE 3 - Reinforced isolation components	Solution 3 (duplicate of above)	•
5	Controller PCBA	Choice of DC/DC converter vs pulse transformer.	Pulse transformer will be used. Significant savings in cost and size.	Implemented	•
6	Controller PCBA	Choice of 12V or 5V DC/DC for AFE power	Replace the 12V/5V DC/DC with a 12V buck to 6V, install a 5V linear regulator on AFE, and drive ovenisation from 6V and AFE control circuitry off 5V.	Implemented	•
7	Controller PCBA	Power output of DC/DC converter is too low to drive ovenisation on AFE board to sufficient capacity.	Increase power to ovenisation when changing to pulse transformer.	2 pulse transformers in parallel gives up to 12W available.	•
8	Controller PCBA	D5 (diode to Super cap) doesn't meet expected pin out. Pin 2 appears to be NC.	Connect cathode to pin 3		•
9	Controller PCBA	Not enough component space as a single sided board for creepage isolation (safety) requirements	Make controller PCB double sided?	Single sided	•
10	Controller PCBA / TSO	Not considered industry best practice for a security device to not include a form of hardware security module. Cannot find level on any of the datasheets.	Add a Infineon or Microchip TPM 1.2 specification SPI /I2C IC with FIPS-140-2 level 2 minimum to both the DC meter controller and FPGA interceptor board	TPM: SLB9670 added	•
11	Controller PCBA	No silkscreen showing AFE isolation barrier	Add silkscreen to show isolation barrier	Determined it was not needed	•
12	Controller PCBA	Centre length board cutout no longer required for positioning in enclosure	Remove board cutout and replace with a mounting point		•
13	Controller PCBA	Designators missing from top overlay for J1-3	Add designators for J1-3 to top overlay	No action.	•
14	Controller PCBA	Sockets J1-3 are positioned too far inwards for proper connection to harnesses	Move sockets J1-3 to be flush with the edge of the PCB		•
15	Controller PCBA	Unnecessary IR LEDs for voltage and time	Remove IR LEDs for voltage and time. Possibility to remove IR LED completely because test pulse output is only needed for certification purposes. Still needs to be installed incase the DC meter is pulled out of a charger and verified on a bench.	Removed volt/time.	•
16	Controller PCBA	Missing test point for GND	Add test point for GND		•
17	Controller PCBA	Footprint for RJ45 housing (CID: 3371) is incorrect. Pin 7 is unnecessary.	Non issue. Left for compatibility to replacement parts.	Part changed anyway for consistency in company.	•
18	Controller PCBA	AFE SPI SCLK line needs 10k pull down	Add 10k pull down resistor		•
19	Controller PCBA	4Gbit external memory maybe insufficient for data purposes (can only be determined once regulation standards become clearer)	Investigate memory requirements	Sufficient given the information we currently have.	•

20	Controller PCBA	J6 and J7 thermistor connectors apparently are not easy to use	Check if these connectors work as expected. If not, replace with more reliable connectors.	Changed to 2 way pico. Fans are 3 way pico	•
21	Controller PCBA	Installed TM4C micro is currently rated 85'C	Purchase industrial version TM4C to ensure temperature range 105'C		•
22	Controller PCBA	No failsafe control if ovenisation is too hot	Micro can turn off isolated power if too hot.		•
23	Controller PCBA	100 MHz noise on negative connection points on both sides of RECOM DC/DC converter	Attach 2K at 100MHz beads on both positive and negative connection points on both input and output of DC/DC converter. Use Richard's RedExpert wondertool to decide on make and model of bead.	Removing Recom DC/DC converter, need to redesign filter.	•
24	Controller PCBA	Decide on new power supply options to replace CAN input power. Calculate power draw of new cooling systems? 12V or 24V input or wide range? Regulated or unregulated?	12V input. No power supply needed. 24V is not an option.	24V12V DCDC brick can always be added externally	•
25	Controller PCBA	Keep KK connectors for ambient and shunt thermisters on controller board?	Change connector type, keep measurement of both ambient and shunt, Richard to decide on which controller board placement side.	Ambient sensor will be onboard.	•
26	Controller PCBA	Debug feature: Software developers need to do timing verification of MCU Require 4xGPIO	Can use the existing debug header. SPI pins can be used as GPIO.		•
27	Controller PCBA	Remove the debug header?	No. Will be kept for GPIO. Configured whichever way firmware is configured.		•
28	Controller PCBA	Is a mounting post required on the long edge?	Yes - We will give location of mounting post to Bernie after placement of both controller and AFE boards.		•
29	Controller PCBA	Location of kW IR LED?	Depends on mechanical design by Bernie	Right edge. Can be vertical or right angle	•
30	Controller PCBA	Fans need to be (on/off) controlled by DC Meter	Add 2x1W Fan outputs.		•
31	AFE PCBA	ADC footprint is not the correct size (too small for actual size of pads). Thermal pad shorting to some surrounding pads.	Increase footprint size to be correct	22 Mar 2019 | A KR | Footprint corrected. Component/PCB needs to be updated	•
32	AFE PCBA	Testpoint for analogue ground near reference voltage location is not ideal. It was not re-located properly after changing ovenisation circuit for fabrication.	Move to better location		•
33	AFE PCBA	Ovenisation MOSFET footprint seems to be oversized for the part. It does work though.	Determine if footprint should be changed	Keeping as is	•
34	AFE PCBA	Socket J1 (maybe) positioned to far inwards for proper connection to harness	Move socket to be flush with PCB edge	22 Mar 2019 | A KR | Harness connects ok as is. Need to investigate tamper prevention requirements before any change made	•
35	AFE PCBA	Designator missing from top (*cough* bottom) overlay for J1	Add designator for J1 to top (*cough* bottom) overlay	22 Mar 2019 | A KR | NC - No designators are included on silk	•
36	AFE PCBA	T-section of the PCB near the mounting post could snap if uneven torque is applied when mounting shunt to busbars	Change AFE board shape on on the T-section past the solder points to the shunt to be narrower to make the AFE board physically more flexible to an unlevel shunt mounting	To be done by DRE after PCB size and shape requirements are finalised.	•
37	AFE PCBA	Clearance distance between C13 (controller board) and AFE components is less than minimal safety requirements	Increase board cutout for capacitor, or install Formex around capacitor	22 Mar 2019 | A KR | Already decided during PCB review to install Formex around capacitor	•
38	AFE PCBA	Which charge cable being used of a Dual PK needs to be determined.	Install a rough voltage sensor, PCB mount connector, optocoupler, trigger IC, and a 4 kV isolation barrier on the AFE.	Voltage sense impletented using comparator (Not isolated)	•
39	AFE PCBA	Temperature control of the ovenisation circuit could fail and cause a fire hazard.	Add an analog switch/relay to the AFE power supply.	Will be on controller board.	•

40	AFE PCBA	Kelvin sense points on future charge cables will not be able to be measured with current AFE design using a 4 channel ADC	Change 4 channel ADS1220 ADC for an 8 channel ADS1278	Scope creep - Decision made to stay with a 4 channel ADC and not include future proofing for Kelvin sense points. Power loss in charge cable will be calculated. DC meter will need to be configured to know which make & model of charge cable and its length is being used.	•
41	AFE PCBA	Analog vs Digital control signal for AFE	Error at 4 A at 25 degC is 2% with ovenisation. Can extrapolate data to generate calibration value. Can't use digital control signal because calibration will be more difficult.	Will remain analogue control. Have kept digital control hardware	•
42	AFE PCBA	Heat retention around AFE PCBA. Put foam inside the canister area, over the can, or over the whole board. Foam material?	Do additional below zero tests with foam over canister and over board. Repeat tests and check for major variances.	With 6W can get to 100'C from -35'C. Need to confirm foam to retain heat	•
43	AFE PCBA	At what power level is the ovenisation MOSFET switching on/off? We want to know whether there is a hard turn offs /ons at high power levels which could damage electronics. Get ADC upto operating temperature, then push it past this temperature and get a capture of power usage/switching signal.	Switching is reasonably slow but can be slowed with the RC on the gate.	Increased R1k. Can be tuned further if necessary	•
44	Both PCBAs	Overlay text for circuit title and company name uses Sans Serif font and looks like it's been written for a children's book and not an industrial application. Could create misunderstandings by technicians when reading (e.g. An I dashed with another I, I-I, could be interpreted as a H.)	`Use monospaced UPPERCASE Serif font for clear readability on all PCBs`	No action	•
45	Shunt	Temperature of shunt at 420 A reaches 85 °C after 25 minutes (and was still rising). 3D printed bottom part partially melted and warped (injection moulded plastic will have a different melting point but will still most likely melt in current design).	Need to investigate whether using a shunt is suitable in the first place because if it produces too much heat and consumes more power than allowed under B. 5.5.2 VDE AR E 2418 3 100 draft E. May need to look into using a fluxgate magnetometer type device. A Hall effect sensor is probably too inaccurate and will be very sensitive to the temperature of the metal strip inside the sensor (i.e. needs constant temperature). If a shunt is still to be used then a heatsink and/or fan will be required.	Decision made to stay with a shunt based current transducer for time being.	•
46	Enclosure	Ability to plug headers to 6 pin micro-molex and RJ45 sockets is restricted	Increase recess to allow better ability to unclip headers		•
47	Enclosure	AGND contact soldered to top of shunt protrudes into wall of top part enclosure	Cutout section around busbars will be expanded to increase airflow for heat dissipation		•
48	Enclosure	Weight of whole DC meter is supported by the shunt being sandwiched between the top and bottom enclosure parts. If enclosure cutout around the shunt is increased to allow additional ventilation air flow, the total weight of the DC meter will be supported only by the solder connection between the shunt contacts and AFE board. Solder contacts could crack or AFE board could snap if uneven torque applied when fastening the two studs in the shunt to busbars.	Investigate an external mounting frame option for the DC meter enclosure to off load the weight being put onto the shunt (could defeat the purpose of a universal shape enclosure for ISD/PK/RTD2 chargers if not done properly.)	Being designed by Bernie	•
49	Request	Provision of Diagnostic LEDs - right now there aren't any These are super handy during development .. also for fault diagnosis in the field If price/view ability is an issue in the field alternatively consider only populating development/early production boards.	Consider the provision of 4 LEDs Heartbeat (Green) - Flashes at 1s frequency when the Meter software is running as expected Out Of Service (Red) - Lit when the Meter is Out Of Service (for whatever reason) Remote Connection (Green) - Lit when connected to remote server (Transparency ??) Charging (Green) - Lit when the Meter considers the Charger is charging	Decision made to not be implemented. 4 debug IO available on spare header.	•
50	Flammability	Fire standards: UL 94 V2 for non live parts (e.g. plastic mounts for PCBS); UL 94 V0 for anything touching live parts.	Use V0 mounts around shunt. V2 everywhere else.		•
51	Ethernet Cables	Connecting shielded ethernet cables to an unearthed device is useless. Does the DC meter need an Earth tap? Connecting shielded would require changing the RJ45 jack.	Use shielded CAT5 cables and replace the RJ45 jack to a shielded jack. Connect shield pin to Bobsmith termination.	Made earth input possible, connected to shield.	•
52	CAN or Ethernet	CAN connectors consumes a lot of PCB surface area	Decision made to remove CAN and install a separate power connection.		•
53	FPGA Carrier PCBA	Incorrect capacitor was purchased for C106 (CID 8663). The footprint on PCB is 1206 which is correct but a 1210 part was purchased during the prototype run.	Nothing. 1210 part would still fit on the 1206 footprint.	Purchase correct part in next rev.	•
54	FPGA Interceptor PCBA	Silkscreen on J4 - P5/N5 is reversed.	N/A	Correct in next rev.	•

55	FPGA Interceptor PCBA	Replace R63 and R64 with a tact switches.	Currently they are DNP and resistor pads need to be shorted to invoke reset.	Correct in next rev.	•
56	FPGA Interceptor PCBA	Leave the 4 Vref pins as GPIOs- floating or tie to 1V. They are now tied to 2.5V VCCO which is incorrect.	Currently this is fixed in the FPGA design to generate internal 0.9V and pins are configured as GPIO.	Correct in next rev.	•
57	FPGA Interceptor PCBA	1.25V Voltage reference U2 chip is incorrectly connected in schematic.	Flip U2 chip around and soldered it to the right pads.	Correct in next rev.	•
58	FPGA Interceptor PCBA	Change R18 to 300 ohm CID 11475 for brighter LED illumination	Change over R18 from 2.2K to 330 ohm.	Correct in next rev.	•
59	FPGA Interceptor PCBA	Use larger pads for PMIC ADP5014. Library footprint needs to be updated.	N/A.	Correct in next rev.	•
60	FPGA Interceptor PCBA	Change R51 to 4.7k ohm resistor CID11468	Change over 10K resistor to 4.7K from stronger pull up.	Correct in next rev.	•
61	FPGA Interceptor PCBA	Add clock source 100MHz to MRCC pin of FPGA	• Possibly use the FPGA internal oscillator • Possibly use the EMCCLK • Possibly use LVDS differential clk • Green wire the ASEMB-100.000MHZ-LY-T to a FPGA clock capable pin	Include clock source ASEMB-100.000MHZ-LY-T in next rev.	•